

Latest Syllabus

PHYSICS (52) CLASS 10

There will be one paper of **two hours** duration carrying 80 marks and Internal Assessment of practical work carrying 20 marks

Note: Unless otherwise specified, only SI Units are to be used while teaching and learning, as well as for answering questions.

1. Force, Work, Power and Energy

- (i) Turning forces concept; moment of a force; forces in equilibrium; centre of gravity; [discussions using simple examples and simple numerical problems].

Elementary introduction of translational and rotational motions; moment (turning effect) of a force, also called torque and its cgs and SI units; common examples - door, steering wheel, bicycle pedal, etc.; clockwise and anti-clockwise moments; conditions for a body to be in equilibrium (translational and rotational); principle of moment and its verification using a metre rule suspended by two spring balances with slotted weights hanging from it; simple numerical problems; Centre of gravity (qualitative only) with examples of some regular bodies and irregular lamina.

- (ii) Uniform circular motion.

As an example of constant speed, though acceleration (force) is present. Differences between centrifugal and centripetal force.

- (iii) Work, energy, power and their relation with force.

Definition of work. $W = FS\cos\theta$; special cases of $\theta = 0^\circ, 90^\circ$. $W = mgh$. Definition of energy, energy as work done. Various units of work and energy and their relation with SI units. [erg, calorie, kW h and eV]. Definition of Power, $P = W/t$; SI and cgs units; other units, kilowatt (kW), megawatt (MW) and gigawatt (GW); and horsepower ($1\text{hp} = 746\text{W}$) [Simple numerical problems on work, power and energy].

- (iv) Different types of energy (e.g., chemical energy, Mechanical energy, heat energy, electrical energy, nuclear energy, sound energy, light energy).

Mechanical energy: potential energy $U = mgh$ (derivation included) gravitational PE, examples; kinetic energy $K = \frac{1}{2}mv^2$ (derivation included);

forms of kinetic energy: translational, rotational and vibrational - only simple examples. [Numerical problems on K and U only in case of translational motion]; qualitative discussions of electrical, chemical, heat, nuclear, light and sound energy, conversion from one form to another; common examples.

- (v) Machines as force multipliers; load, effort, mechanical advantage, velocity ratio and efficiency; simple treatment of levers, pulley systems showing the utility of each type of machine.

Functions and uses of simple machines: Terms- effort E, load L, mechanical advantage $MA = L/E$, velocity ratio $VR = V_E/V_L = d_E/d_L$, input (W_i), output (W_o), efficiency (η), relation between η and MA, VR (derivation included); for all practical machines $\eta < 1$; $MA < VR$.

Lever: principle. First, second and third class of levers; examples: MA and VR in each case. Examples of each of these classes of levers as also found in the human body.

Pulley system: single fixed, single movable, block and tackle; MA, VR and η in each case.

- (vi) Principle of Conservation of energy.

Statement of the principle of conservation of energy; theoretical verification that $U + K = \text{constant}$ for a freely falling body. Application of this law to simple pendulum (qualitative only); [simple numerical problems].

2. Light

- (i) Refraction of light through a glass block and a triangular prism - qualitative treatment of simple applications such as real and apparent depth of objects in water and apparent bending of sticks in water. Applications of refraction of light.

Partial reflection and refraction due to change in medium. Laws of refraction; the effect on speed (V), wavelength (λ) and frequency (f) due to refraction of light; conditions for a light ray to pass undeviated. Values of speed of light (c) in vacuum, air, water and

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glass; refractive index $\mu = c/V$, $V = f\lambda$. Values of μ for common substances such as water, glass and diamond; experimental verification; refraction through glass block; lateral displacement; multiple images in thick glass plate/mirror; refraction through a glass prism, simple applications: real and apparent depth of objects in water; apparent bending of a stick under water. (Simple numerical problems and approximate ray diagrams required).

- (ii) Total internal reflection: Critical angle; examples in triangular glass prisms; comparison with reflection from a plane mirror (qualitative only). Applications of total internal reflection.

Transmission of light from a denser medium (glass/water) to a rarer medium (air) at different angles of incidence; critical angle (C) $\mu = 1/\sin C$. Essential conditions for total internal reflection. Total internal reflection in a triangular glass prism; ray diagram, different cases - angles of prism ($60^\circ, 60^\circ, 60^\circ$), ($60^\circ, 30^\circ, 90^\circ$), ($45^\circ, 45^\circ, 90^\circ$); use of right angle prism to obtain $\delta = 90^\circ$ and 180° (ray diagram); comparison of total internal reflection from a prism and reflection from a plane mirror.

- (iii) Lenses (converging and diverging) including characteristics of the images formed (using ray diagrams only); magnifying glass; location of images using ray diagrams and thereby determining magnification.

Types of lenses (converging and diverging), convex and concave, action of a lens as a set of prisms; technical terms; centre of curvature, radii of curvature, principal axis, foci, focal plane and focal length; detailed study of refraction of light in spherical lenses through ray diagrams; formation of images - principal rays or construction rays; location of images from ray diagram for various positions of a small linear object on the principal axis; characteristics of images. Sign convention and direct numerical problems using the lens formula are included (derivation of formula not required).

Scale drawing or graphical representation of ray diagrams not required.

Power of a lens (concave and convex) – [simple direct numerical problems]; magnifying glass or simple microscope: location of image and magnification from ray diagram only [formula and numerical problems not included]. Applications of lenses.

- (iv) Using a triangular prism to produce a visible spectrum from white light; Electromagnetic spectrum. Scattering of light.

Deviation produced by a triangular prism; dependence on colour (wavelength) of light; dispersion and spectrum; electromagnetic spectrum: broad classification (names only arranged in order of increasing wavelength); properties common to all electromagnetic radiations; properties and uses of infrared and ultraviolet radiation. Simple application of scattering of light e.g. blue colour of the sky.

3. Sound

- (i) Reflection of Sound Waves; echoes: their use; simple numerical problems on echoes.

Production of echoes, condition for formation of echoes; simple numerical problems; use of echoes by bats, dolphins, fishermen, medical field. SONAR.

- (ii) Natural vibrations, Damped vibrations, Forced vibrations and Resonance - a special case of forced vibrations.

Meaning and simple applications of natural, damped, forced vibrations and resonance.

- (iii) Loudness, pitch and quality of sound:

Characteristics of sound: loudness and intensity; subjective and objective nature of these properties; sound level in decibel (dB) (as unit only); noise pollution; interdependence of: pitch and frequency; quality and waveforms (with examples).

4. Electricity and Magnetism

- (i) Ohm's Law; concepts of emf, potential difference, resistance; resistances in series and parallel, internal resistance.

Concepts of pd (V), current (I), resistance (R) and charge (Q). Ohm's law: statement, $V=IR$; SI units; experimental verification; graph of V vs I and resistance from slope; ohmic and non-ohmic resistors, factors affecting resistance (including specific resistance) and internal resistance; super conductors, electromotive force (emf); combination of resistances in series and parallel and derivation of expressions for equivalent resistance. Simple numerical problems using the above relations. [Simple network of resistors].

- (ii) Electrical power and energy.

Electrical energy; examples of heater, motor, lamp, loudspeaker, etc. Electrical power; measurement of electrical energy, $W = QV = VIt$ from the definition

of pd. Combining with ohm's law $W = VIt = I^2 Rt = (V^2/R)t$ and electrical power $P = (W/t) = VI = I^2 R = V^2/R$. Units: SI and commercial; Power rating of common appliances, household consumption of electric energy; calculation of total energy consumed by electrical appliances; $W = Pt$ (kilowatt $\times$ hour = kWh), [simple numerical problems].

- (iii) Household circuits – main circuit; switches; fuses; earthing; safety precautions; three-pin plugs; colour coding of wires.

House wiring (ring system), power distribution; main circuit (3 wires-live, neutral, earth) with fuse / MCB, main switch and its advantages - circuit diagram; two-way switch, staircase wiring, need for earthing, fuse, 3-pin plug and socket; Conventional location of live, neutral and earth points in 3 pin plugs and sockets. Safety precautions, colour coding of wires.

- (iv) Magnetic effect of a current (principles only, laws not required); electromagnetic induction (elementary); transformer.

Oersted's experiment on the magnetic effect of electric current; magnetic field (B) and field lines due to current in a straight wire (qualitative only), right hand thumb rule – magnetic field due to a current in a loop; Electromagnets: their uses; comparisons with a permanent magnet; Fleming's Left Hand Rule, the DC electric motor- simple sketch of main parts (coil, magnet, split ring commutators and brushes); brief description and type of energy transfer(working not required): Simple introduction to electromagnetic induction; frequency of AC in house hold supplies, Fleming's Right Hand Rule, AC Generator - Simple sketch of main parts, brief description and type of energy transfer(working not required). Advantage of AC over DC. Transformer- its types, characteristics of primary and secondary coils in each type (simple labelled diagram and its uses).

5. Heat

- (i) Calorimetry: meaning, specific heat capacity; principle of method of mixtures; Numerical Problems on specific heat capacity using heat loss and gain and the method of mixtures.

Heat and its units (calorie, joule), temperature and its units ($^{\circ}C$, K); thermal (heat) capacity $C = Q/\Delta T$.. (SI unit of C): Specific heat Capacity $C = Q/m\Delta T$ (SI unit of C) Mutual relation between Heat Capacity and Specific Heat capacity, values of C for some common substances (ice, water and copper). Principle of method of mixtures including mathematical statement. Natural phenomenon involving specific

heat. Consequences of high specific heat of water. [Simple numerical problems].

- (ii) Latent heat; loss and gain of heat involving change of state for fusion only.

Change of phase (state); heating curve for water; latent heat; specific latent heat of fusion (SI unit). Simple numerical problems. Common physical phenomena involving latent heat of fusion.

6. Modern Physics

- (i) Radioactivity and changes in the nucleus; background radiation and safety precautions.

Brief introduction (qualitative only) of the nucleus, nuclear structure, atomic number (Z), mass number (A). Radioactivity as spontaneous disintegration. α , β and γ - their nature and properties; changes within the nucleus. One example each of α and β decay with equations showing changes in Z and A. Uses of radioactivity - radio isotopes. Harmful effects. Safety precautions. Background radiation.

Radiation: X-rays; radioactive fallout from nuclear plants and other sources.

Nuclear Energy: working on safe disposal of waste. Safety measures to be strictly reinforced.

- (ii) Nuclear fission and fusion; basic introduction and equations.

A NOTE ON SI UNITS

SI units (Système International d'Unités) were adopted internationally in 1968.

Fundamental units

The system has seven fundamental (or basic) units, one for each of the fundamental quantities.

Fundamental quantity	Unit	
	Name	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Luminous intensity	candela	cd
Amount of substance	mole	mol

Derived units

These are obtained from the fundamental units by multiplication or division; no numerical factors are involved. Some derived units with complex names are:

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Derived quantity	Unit	
	Name	Symbol
Volume	cubic metre	m^3
Density	kilogram per cubic metre	kg m^{-3}
Velocity	metre per second	m s^{-1}
Acceleration	metre per second square	m s^{-2}
Momentum	kilogram metre per second	kg m s^{-1}

Some derived units are given special names due to their complexity when expressed in terms of the fundamental units, as below:

Derived quantity	Unit	
	Name	Symbol
Force	newton	N
Pressure	pascal	Pa
Energy, Work	joule	J
Power	watt	W
Frequency	hertz	Hz
Electric charge	coulomb	C
Electric resistance	ohm	Ω
Electromotive force	volt	V

When the unit is named after a person, the symbol has a capital letter.

Standard prefixes

Decimal multiples and submultiples are attached to units when appropriate, as below:

Multiple	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f

INTERNAL ASSESSMENT OF PRACTICAL WORK

Candidates will be asked to carry out experiments for which instructions will be given. The experiments may be based on topics that are not included in the syllabus but theoretical knowledge will not be required. A candidate will be expected to be able to follow simple instructions, to take suitable readings and to present these readings in a systematic form. He/she may be required to exhibit his/her data graphically. Candidates will be expected to appreciate and use the concepts of least count, significant figures and elementary error handling.

Note: Teachers may design their own set of experiments, preferably related to the theory syllabus. A comprehensive list is suggested below:

1. Lever - There are many possibilities with a meter rule as a lever with a load (known or unknown) suspended from a point near one end (say left), the lever itself pivoted on a knife edge, use slotted weights suspended from the other (right) side for effort.
Determine the mass of a metre rule using a spring balance or by balancing it on a knife edge at some point away from the middle and a 50g weight on the other side. Next pivot (F) the metre rule at the 40cm, 50cm and 60cm mark, each time suspending a load L or the left end and effort E near the right end. Adjust E and or its position so that the rule is balanced. Tabulate the position of L, F and E and the magnitudes of L and E and the distances of load arm and effort arm. Calculate $MA = L/E$ and $VR = \text{effort arm/load arm}$. It will be found that $MA < VR$ in one case, $MA = VR$ in another and $MA > VR$ in the third case. Try to explain why this is so. Also try to calculate the real load and real effort in these cases.
2. Determine the VR and MA of a given pulley system.
3. Trace the course of different rays of light refracting through a rectangular glass slab at different angles of incidence, measure the angles of incidence, refraction and emergence. Also measure the lateral displacement.
4. Determine the focal length of a convex lens by (a) the distant object method and (b) using a needle and a plane mirror.
5. Determine the focal length of a convex lens by using two pins and formula $f = uv/(u+v)$.

6. For a triangular prism, trace the course of rays passing through it, measure angles i_1 , i_2 , A and δ . Repeat for four different angles of incidence (say $i_1 = 40^\circ$, 50° , 60° and 70°). Verify $i_1 + i_2 = A + \delta$ and $A = r_1 + r_2$.
7. For a ray of light incident normally ($i_1 = 0$) on one face of a prism, trace course of the ray. Measure the angle δ . Explain briefly. Do this for prisms with $A = 60^\circ$, 45° and 90° .
8. Calculate the specific heat capacity of the material of the given calorimeter, from the temperature readings and masses of cold water, warm water and its mixture taken in the calorimeter.
9. Determination of specific heat capacity of a metal by method of mixtures.
10. Determination of specific latent heat of ice.
11. Using as simple electric circuit, verify Ohm's law. Draw a graph, and obtain the slope.
12. Set up model of household wiring including ring main circuit. Study the function of switches and fuses.

Teachers may feel free to alter or add to the above list. The students may perform about ten experiments. Some experiments may be demonstrated.

EVALUATION

The practical work/project work are to be evaluated by the subject teacher and by an External Examiner. (The External Examiner may be a teacher nominated by the Head of the school, who could be from the faculty, **but not teaching the subject in the relevant section/class**. For example, a teacher of Physics of Class VIII may be deputed to be an External Examiner for Class X, Physics projects.)

The Internal Examiner and the External Examiner will assess the practical work/project work independently.

Award of Marks (20 Marks)

Subject Teacher (Internal Examiner)	10 marks
External Examiner	10 marks

The total marks obtained out of 20 are to be sent to ISCE by the Head of the school.

The Head of the school will be responsible for the online entry of marks on CISCE's CAREERS portal by the due date.

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- Sum of clockwise moments about a point (or axis) must be equal to the sum of anticlockwise moments about the same point.

$$\Sigma \tau_{\text{clockwise}} = \Sigma \tau_{\text{anticlockwise}}$$

5. Principle of moment

- Measures the turning effect of a force applied to a body.

$$\text{Torque } (\tau) = \hat{F} \times \hat{r}$$

SI unit: newton-metre (Nm)

1. Moment of Force (Torque)

Force

2. Conditions for Equilibrium

• Translational Equilibrium

A body is in translational equilibrium when the net force acting on it is zero.

$$\Sigma F_x = 0 \text{ and } \Sigma F_y = 0$$

• Rotational Equilibrium

A body is in rotational equilibrium when the net torque acting on it is zero.

$$\Sigma \tau = 0$$

3. Centre of gravity

- The point at which the entire weight of the body can be considered to act, regardless of the orientation of the body.
- For a body of **uniform shape and density**, center of gravity coincides with geometric center.
- For an **irregular-shaped body**, the CG may lie the outside physical boundaries of body.

- Motion of an object traveling in a circular path at a constant speed.

Centripetal Acceleration:

Acceleration directed toward centre of circle:

$$a_c = \frac{v^2}{r}$$

Centripetal Force"

Force responsible for maintaining circular motion,

acting toward center of circle : $F_c = \frac{mv^2}{r}$

Centrifugal force :

It acts outward from centre of the circle and is equal in magnitude but opposite in direction to centripetal force.

4. Uniform circular motion

• **Kinetic Energy (KE):** Energy due to motion

Types

1. **Translational K.E:** Energy due to the motion of an object moving in a straight line or along a path [$KE = \frac{1}{2}mv^2$]

2. **Rotational K.E:** Energy due to the rotation of an object about an axis.

3. **Vibrational K.E:** Energy due to the vibration of particles in an object or system.

• **Potential Energy (PE):** Energy due to position or configuration

Types

1. **Gravitational P.E:** Energy stored in an object due to its position in a gravitational field. [$U = mgh$]

2. **Elastic P.E:** Energy stored in objects that can be stretched, compressed, or deformed, like springs or rubber bands.

Type of Energy	Key Source/Example	Applications
Mechanical Energy	Moving or positioned objects	Vehicles, turbines
Chemical Energy	Fuel, batteries, food	Combustion, metabolic processes
Thermal Energy	Heat from combustion or friction	Heating, steam turbines
Electrical Energy	Flow of electrons	Appliances, circuits
Nuclear Energy	Nuclear reactions	Power plants, weapons
Light Energy	Sunlight, artificial lights	Solar panels, vision
Sound Energy	Vibrating objects	Music, communication
Elastic Energy	Stretched or compressed materials	Springs, bows
Gravitational Energy	Height of position in gravitational fields	Hydropower, stored water
Magnetic Energy	Magnetic fields	Generators, motors

• It states that energy can neither be created nor destroyed;

it can only be transformed from one form to another or transferred between systems.

The total energy of an isolated system remains constant over time.

Conversion	Example
Potential Energy → Kinetic Energy	Pendulum, roller coaster
Chemical Energy → Heat Energy	Burning fuel, cooking
Chemical Energy → Electrical Energy	Batteries, fuel cells
Electrical Energy → Light Energy	Bulbs, LEDs
Electrical Energy → Mechanical Energy	Fans, electric motors
Mechanical Energy → Electrical Energy	Wind turbines, hydroelectric power plants
Nuclear Energy → Electrical Energy	Nuclear power plants
Solar Energy → Electrical Energy	Solar panels
Heat Energy → Mechanical Energy	Steam engines, car engines
Sound Energy → Electrical Energy	Microphones
Light Energy → Chemical Energy	Photosynthesis, solar cells
Mechanical Energy → Heat Energy	Brakes, frictional heat

5. Different types of energy

Work, energy, power

2. Power

3. Energy

• Energy is the capacity to do work : $Energy \sim Work\ done$

• Work is done when a force is applied to an object and object moves in the direction of the applied force.

$$W = \vec{F} \cdot \vec{S} = FS\cos\theta$$

Where: W = work done (in joules, J), F = force applied (in newtons, N),

S = displacement of the object (in metre, m), θ = angle between force and direction of displacement.

Maximum work (Positive work): $\theta = 0^\circ$

No work done (Zero work): $\theta = 90^\circ$

Minimum work (Negative work): $\theta = 180^\circ$

• It is defined as the rate at which work is done or energy is transferred.

$$P = \frac{W}{t} = \frac{E}{t}$$

Where: P = Power (in watts), W = Work done (in joules),

E = Energy (in joules), t = time (in seconds)

Mechanical Power: $P = Fv$

Electrical Power: $P = VI$

where: P = Power (in watts, W),

F = Force (N), v = Velocity (m/s),

V = Potential difference (V), I = Current (A).

Unit	Relation to Watts (W)
Watt (W)	1 W
Erg/second	1 W = 10^7 erg/s
Kilowatt (kW)	1 kW = 1000 W
Megawatt (MW)	1 MW = 10^6 W
Gigawatt (GW)	1 GW = 10^9 W
Horsepower (hp)	1 hp = 746 W

Unit	Definition	Relation to Joules (J)
Erg	The amount of work done by a force of one dyne exerted for a distance of one centimetre.	1 erg = 10^{-7} J
Calorie	The amount of heat required to raise the temperature of one gram of water by one degree Celsius.	1 cal = 4.186 J
Kilowatt-hour	Energy consumed by 1 kW over 1 hour.	1 kWh = 3.6×10^6 J
Electron-volt	The amount of kinetic energy gained by a single electron when it is accelerated through an electric potential difference of one volt.	1 eV = 1.602×10^{-19} J

- **Equilateral Prism:** Used for dispersion of light into its constituent colors due to variation of refractive indices for different colours.
- **Right-Angled Prism:** Used in periscopes and binoculars to turn light paths by 90° Used for specific optical beam manipulation.

Comparison of Total Internal Reflection from a Prism and Reflection from a Plane Mirror

Aspect	TIR from a Prism	Reflection from a Plane Mirror
Efficiency	Nearly 100% reflection (no energy loss).	Some energy is lost due to absorption by the reflective coating.
Image Quality	Produces brighter, clearer images.	Images may be slightly dimmer and degrade over time.
Durability	Prisms are robust and last longer.	Mirrors may lose reflectivity over time due to tarnishing.
Angle Control	Precisely controls light paths (e.g., 90° or 180°).	Reflection angle depends entirely on the angle of incidence.
Material	Requires transparent material (glass or similar).	Requires a reflective coating on a plane surface.

5. TIR in Triangular Glass Prism

• Mirage Formation:

In deserts, light from the sky undergoes TIR at the rarer layers of hot air near the ground, creating an illusion of water.

Optical Fibers:

Light is confined within the fiber core by repeated TIR from cladding, enabling high-speed data transmission.

Periscopes and Binoculars:

Right-angled prisms use TIR to change the direction of light efficiently.

- **Total Internal Reflection (TIR)** is the phenomenon in which a light ray traveling from a denser medium to a rarer medium is completely reflected back into the denser medium, instead of refracting into the rarer medium.

• Condition for TIR:

The light travels from a denser medium (e.g., glass or water) to a rarer medium (e.g., air).

The angle of incidence is greater than the critical angle (C) for the pair of media.

$$\text{Formula : } \sin C = \frac{1}{\mu}$$

3. Total internal reflection

Refraction of light

1. Refraction

Effect of Refraction on Speed (v), Wavelength (λ) and Frequency (f)

$$\text{Formula : } v = \lambda f$$

Property	Denser Medium	Rarer Medium
Speed (v)	Decreases	Increases
Wavelength (λ)	Decreases	Increases
Frequency (f)	Remains constant	Remains constant

Refractive Index:

It is the ratio of the speed of light in a vacuum (c) to the speed of light in the medium (v); $\mu = \frac{c}{v}$

$$\text{For any other two media: } \frac{\lambda_2}{\lambda_1} = \frac{v_2}{v_1} = \frac{\mu_1}{\mu_2}$$

• Refraction Through a Glass Block

- At the air-glass interface (entry point): The light ray bends towards the normal as it enters the denser medium (glass).
- At the glass-air interface (exit point): The light ray bends away from the normal as it exits into the rarer medium (air).

$$\text{Lateral Displacement (d): } d = \frac{t \cdot \sin(i - r)}{\cos(r)}$$

where: t = Thickness of the glass block

• Real and Apparent Depth

When light passes from water to air (denser to rarer medium), it bends away from the normal, creating an optical illusion where objects submerged in water appear shallower than their actual depth.

$$\mu = \frac{d_{\text{real}}}{d_{\text{apparent}}}$$

• Apparent Bending of a Stick Under Water

A straight stick partially submerged in water appears bent at the water's surface. This occurs due to refraction at the air-water interface.

2. Applications of refraction of light.

• Laws of Refraction

First Law (Snell's Law): The ratio of the sine of the angle of incidence (i) to the sine of the angle of refraction (r) is constant for a given pair of media.

$$\frac{\sin i}{\sin r} = \mu_{21} = \frac{v_1}{v_2}$$

μ_{21} = Refractive index of the second medium with respect to the first.

v_1, v_2 : Speeds of light in the two media.

i = Angle of incidence, r = Angle of refraction.

Second Law: The incident ray, refracted ray and the normal at the point of incidence all lie in the same plane.

• **Refraction** is the phenomenon where the direction of a light changes as it passes obliquely from one medium into another, caused by a change in its speed.

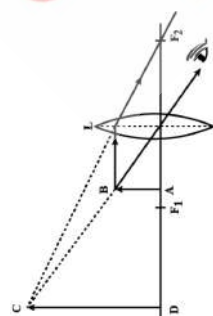
•Convex Lenses:

Magnifying glass, Camera lenses, Telescopes and microscopes

•Concave Lenses:

Used in spectacles for correcting myopia (short-sightedness), Peepholes in doors

Applications of lenses.



•Image Characteristics:

Location: Virtual, on the same side as the object.

Nature: Virtual and upright.

Size: Enlarged.

Simple microscope

- Power of a lens is a measure of its ability to converge or diverge light.

It is defined as the reciprocal of the focal length (f) in meters:

$$P = \frac{1}{f} \text{ (in diopters, D)}$$

Positive power : Convex lenses.

Negative power : Concave lenses.

Power of a lens

- Lens Formula:** Lens formula relates the object distance (u), image distance (v), and focal length (f):

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

- Magnification (M)** : Magnification is the ratio of the height of the image (h_i) to the height of the object (h_o) or the ratio of the image distance to the object distance:

$$M = \frac{v}{u} = \frac{h_i}{h_o}$$

Lens and Magnification formula

- Optical Center as Origin:** The optical center of the lens is taken as the origin, and the principal axis is considered as the X-axis.

- Distances Measured from the Optical Center:**

Distances measured in the direction opposite to the incident light.

Distances measured in the direction of incident light.

- Heights Measured Vertically:**

Heights above the principal axis are positive.

Heights below the principal axis are negative.

- Focal Length:**

For a convex lens, the focal length (f) is positive.

For a concave lens, the focal length (f) is negative.

Sign convention

- Lenses are optical devices made of transparent materials with at least one curved surface.

Types 1.Converging Lens (Convex Lens),

2. Diverging Lens (Concave Lens)

Lens

- Optical Center (O)** : optical center is the central point of the lens through which a ray of light passes without any deviation.

- **Radius of Curvature (R)**:radius of curvature is the radius of the spherical surface of which the lens forms a part.

- Principal Axis** : principal axis is an imaginary straight line that passes through centres of curvature.

- Principal Focus (F)** : • The point on the principal axis, where light rays parallel to the principal axis converge (in case of convex lens) or appear to

diverge from a point (in case of concave lens)

after refraction. A lens has two principal foci named F_1 and F_2 .

- Focal Length (f)** : focal length is the distance between

the optical center (O) and the focus (F) of the lens.

It is positive for convex lenses and negative for concave lenses.

- Aperture:**aperture is the effective diameter of the lens through which light passes.

It determines the amount of light entering the lens.

Terms related to Lenses

Lenses

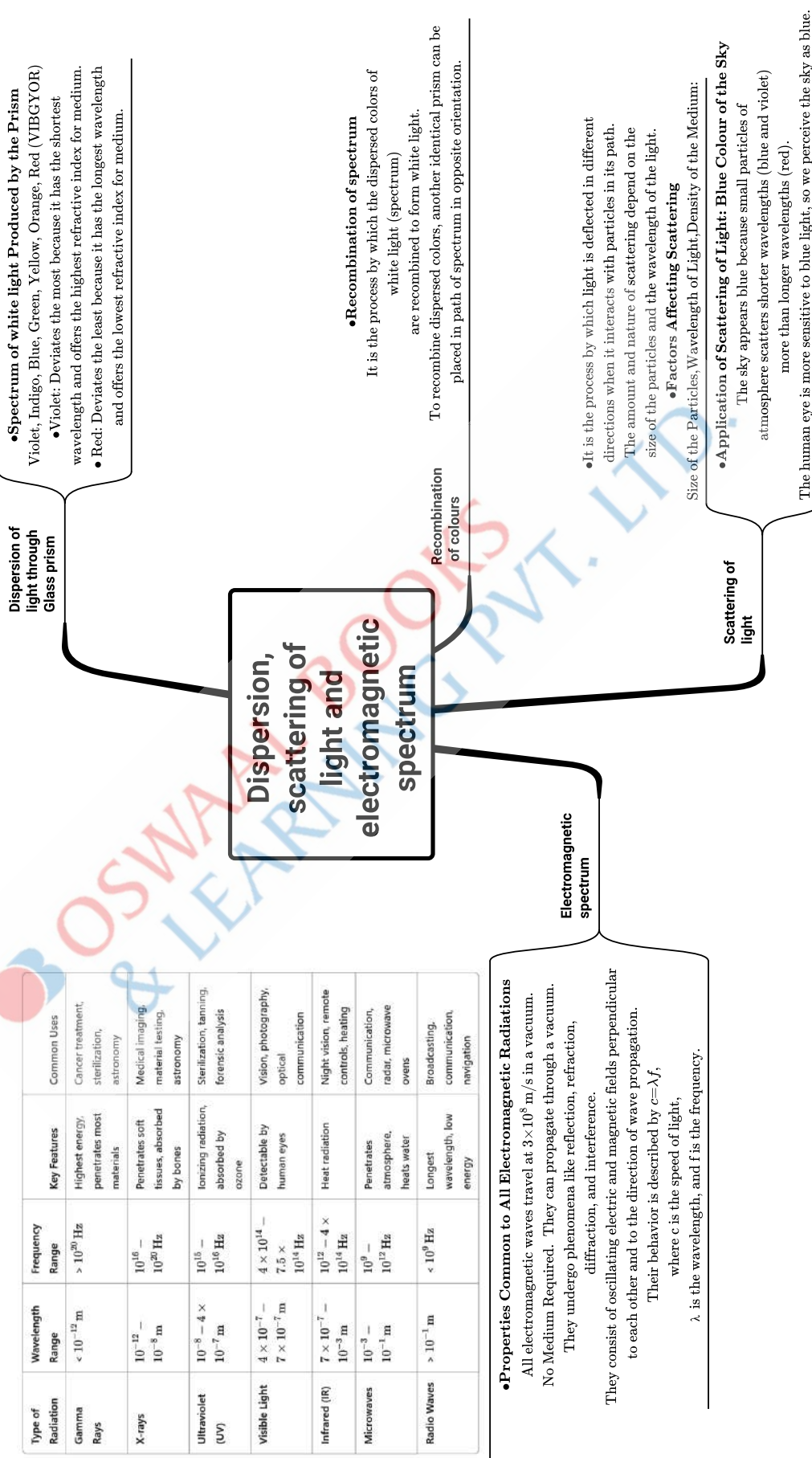
Object Position	Image Location	Nature of Image	Size of Image
At infinity	At focus (F)	Real and inverted	Highly diminished (point-sized)
Beyond 2F	Between F and 2F	Real and inverted	Diminished
At 2F	At 2F	Real and inverted	Same size as the object
Between F and 2F	Beyond 2F	Real and inverted	Enlarged
At F	At infinity	Real and inverted	Highly enlarged
Between F and optical center	Same side as object	Virtual and upright	Enlarged

By Convex Lens

Object Position	Image Location	Nature of Image	Size of Image
At infinity	At focus (F) (same side as object)	Virtual and upright	Highly diminished (point-sized)
Anywhere in front of the lens	Between F and O	Virtual and upright	Diminished

By Concave Lens

Formation of images



- **Potential Difference (V):** work done to move a unit charge from one point to another in an electric circuit

$$V = \frac{W}{Q}; \text{Unit: volt (V): 1 volt} = 1 \text{ joule per coulomb.}$$

Instrument : Measured using a voltmeter, connected in parallel to the component.

- **Electric Current (I):** The flow of electric charge through a conductor per unit time.

$$I = \frac{Q}{t}; \text{Unit: ampere (A): 1 ampere} = 1 \text{ coulomb per second.}$$

Instrument: Measured using an ammeter, connected in series.

- **Resistance (R):** The opposition offered by a material to the flow of electric current.

$$R = \rho \frac{L}{A}; \text{Unit: ohm } (\Omega); 1 \text{ ohm} = 1 \text{ volt per ampere.}$$

Factors Affecting Resistance:

1. Length of the conductor ($R \propto L$), 2. Cross-sectional area ($R \propto \frac{1}{A}$)

3. Material (resistivity, ρ), 4. Temperature (varies for different materials)

A **superconductor** is a material that has zero electrical resistance when cooled below a critical temperature (T_c)

- **Resistivity (ρ):** It is a property of a material that measures its ability to resist flow of electric current.

Unit: ohm-metre ($\Omega \cdot m$)

Low Resistivity: Indicates a good conductor of electricity (e.g., silver, copper).

High Resistivity: Indicates a good insulator (e.g., glass, rubber).

For most conductors, resistivity increases with an increase in temperature.

For semiconductors, resistivity decreases as temperature increases.

- **Conductivity (σ):** It is the measure of a material's ability to allow the flow of electric current

$$\sigma = \frac{1}{\rho}; \text{Unit: siemens per meter (S/m), 1 siemens (S)} = \frac{1}{\Omega}$$

- **Electromotive Force (EMF, ϵ)**

Electromotive Force is the energy supplied per unit charge by a power source, such as a battery or generator.

Unit: volts (V)

- **Internal Resistance (r):**

Internal resistance is the small resistance inside a power source that opposes the flow of current.

Factors Affecting Internal Resistance: Temperature, Electrode Surface Area, Electrolyte Concentration

- **Terminal Potential (V)**

Terminal potential is the actual voltage available across the terminals of a power source when it is connected to a circuit and current is flowing.

$$V = \epsilon - Ir$$

- **Current Formula for EMF**

$$I = \frac{\epsilon}{R+r}, \text{ where: } R = \text{External resistance } (\Omega)$$

1. Combination of Resistors

- **Series Combination** $R_{\text{total}} = R_1 + R_2 + R_3 + \dots + R_n$
- **Parallel Combination** $\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$

Property	Series Combination	Parallel Combination
Current	Same through all resistors.	Divides among the resistors.
Voltage	Divides across resistors.	Same across all resistors.
Total Resistance (R)	Increases with added resistors ($R_{\text{total}} > R_{\text{individual}}$).	Decreases with added resistors ($R_{\text{total}} < R_{\text{individual}}$).
Applications	Used to limit current (e.g., voltage divider).	Used to distribute current (e.g., household wiring).

- **Ohm's Law:** The current (I) flowing through a conductor is directly proportional to the potential difference (V) across its ends, provided the temperature and other physical conditions remain constant.

$$V = I \cdot R$$

Current Electricity

2. Ohm's Law

Property	Ohmic Conductors	Non-Ohmic Conductors
Definition	Materials that obey Ohm's law ($V \propto I$).	Materials that do not obey Ohm's law.
Graph	Straight line through the origin (linear).	Non-linear curve.
Resistance	Constant, does not change with voltage or current.	Varies with voltage or current.
Examples	Metals like copper, aluminum and resistors.	Diodes, transistors, filament bulbs, semiconductors.
Behavior	Current increases uniformly with an increase in voltage.	Current does not increase uniformly with voltage.

- **Ohmic and Non-Ohmic Conductors:**

4. Basic Terms

3. Concepts of EMF

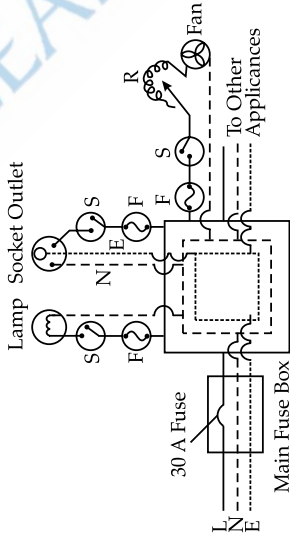
Wire Function	Wire Color (India - IEC Standard)	Used For
Live (L1) (Phase 1)	Red or Brown	Power supply to electrical devices
Neutral (N)	Black or Blue	Completes the circuit
Earth/Ground (E)	Green or Green/Yellow	Safety, prevents shocks

Color code of wire

Electrical Energy

- **Definition :** It is the energy transferred by electric charges as they move through a conductor or circuit.
 $E = V \times I \times t = P \times t$
Where: E = Electrical energy (Joules, J), P = Power (Watts, W)
1 kWh = 3.6×10^6 J

Main Switch—Controls the power supply to the entire house.
MCB (Miniature Circuit Breaker)—Protects circuits from overloading and short circuits.
Fuse—Traditional safety device that breaks the circuit in case of excess current.
Distribution Board (DB) / Fuse Box—Contains MCBs and divides power into different circuits.
Sockets and Switches—Power outlets for appliances.



Ring System of House Wiring

- Reduced Wire Usage**—Requires less copper than radial circuits.
- Higher Load Capacity**—Can handle more appliances safely.
- Less Voltage Drop**—Ensures stable voltage across all sockets.
- Efficient Power Distribution**—Power flows in two directions, reducing voltage drop.
- Backup Power Path**—If one part of the ring is disconnected, electricity still flows from the other side.

- Advantages**
 - Ensure Proper Earthing to avoid electric shocks.
 - Use High Quality Wires to prevent overheating.
 - Keep Wires and Appliances Dry
 - Fix Loose Connections to prevent sparks.
 - Childproof Sockets to avoid accidents.
- Safety precautions**

Electrical energy and Household circuit

Electrical Power

- **Definition :** Electrical power is the rate at which electrical energy is converted into other forms of energy (like heat, light, mechanical energy, etc.) in a circuit.
 $P = V \times I = E/t = I^2 \times R = \frac{V^2}{R}$
Unit: watts (W), 1 hp = 746 W

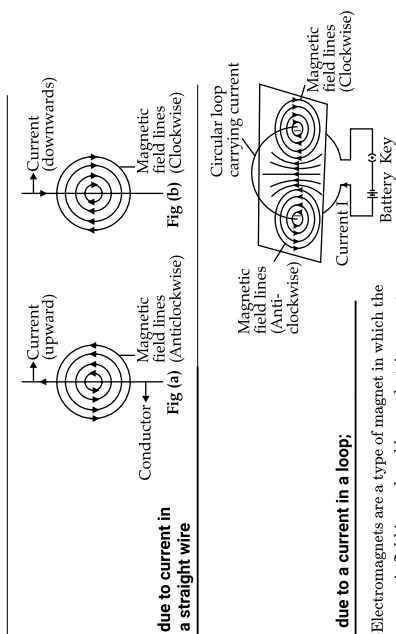
- Hans Christian Oersted (1820) discovered that electric current creates a magnetic field around a conductor.

Magnetic field (B) is measured in **tesla (T)**.

Right-Hand Thumb Rule:

Thumb $\rightarrow$ Direction of current

Fingers curl $\rightarrow$ Direction of magnetic field



1. Magnetic field

due to current in a straight wire

Conductor

Fig (a) (Anticlockwise)

Fig (b) (Clockwise)

Circular loop carrying current

Magnetic field lines (Anti-clockwise)

Magnetic field lines (Clockwise)

Current I

Battery Key

due to a current in a loop;

- Electromagnets are a type of magnet in which the magnetic field is produced by an electric current.

Feature	Electromagnet	Permanent Magnet
Magnetic Field	Can be turned on/off and adjusted.	Always on and fixed.
Control	Field strength and direction are controllable.	Fixed field strength and direction.
Power Source	Requires electricity to work.	No power source needed.
Strength	Adjustable based on current and number of turns in coils.	Fixed strength.
Applications	Motors, relays, lifting, MRI, etc.	Compasses, fridge magnets, etc.

2. Electromagnets

Magnetic effect of a current

3. Fleming's Left Hand Rule

• Thumb: Represents the Force (motion of the conductor).

First Finger: Represents the Magnetic Field (from North to South).

Second Finger: Represents the Current (from positive to negative).

- A DC electric motor is a device that converts electrical energy into mechanical energy through the interaction of a magnetic field and an electric current.

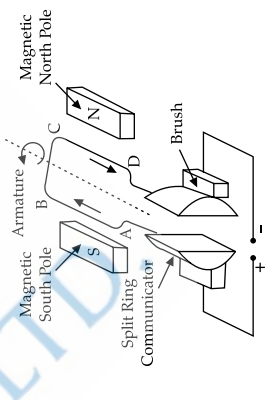
• Coil (Armature): Rotates in the magnetic field and carries current.

Permanent Magnet: Provides the magnetic field.

Split Ring Commutator: Reverses current direction to maintain rotation.

Brushes: Maintain electrical contact with the commutator.

Main parts



4. DC electric motor

8. Transformer

Characteristic	Step-Up Transformer	Step-Down Transformer
Coil Turns (Primary)	Fewer turns	More turns
Coil Turns (Secondary)	More turns	Fewer turns
Voltage (Primary)	Lower	Higher
Voltage (Secondary)	Higher	Lower
Purpose	Increases voltage	Decreases voltage
Use	Power transmission	Household and industrial use

- A transformer is an electrical device that changes the voltage of alternating current (AC) in a circuit. It operates on the principle of electromagnetic induction

Formula: $\frac{V_p}{V_s} = \frac{N_p}{N_s}$

Where: V_p = Primary voltage

V_s = Secondary voltage

N_p = Number of turns in the primary coil

N_s = Number of turns in the secondary coil

- An AC generator converts mechanical energy into electrical energy by using the principle of electromagnetic induction.

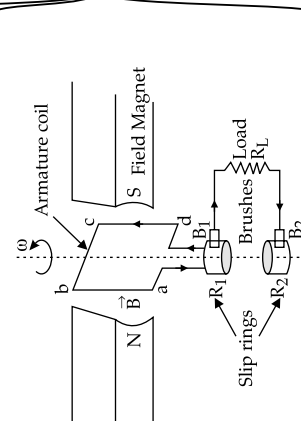
• Coil (Armature): Rotates within the magnetic field, generating an alternating current.

Magnet: Provides a magnetic field in which the coil rotates.

Slip Rings: Allow continuous connection to the rotating coil.

Brushes: Maintain electrical contact with the slip rings, transferring the generated current to the external circuit.

Main parts



• Thumb: Direction of motion (of the conductor).

First Finger: Direction of magnetic field (from North to South).

Second Finger: Direction of induced current (from positive to negative).

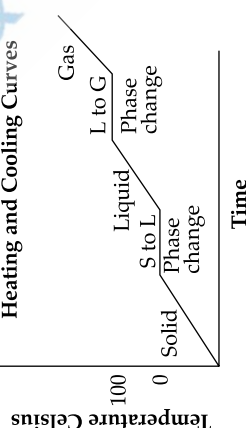
6. Fleming's Right Hand Rule

- Electromagnetic induction is the process by which a changing magnetic field induces an electric current in a conductor

5. Electromagnetic Induction

Phase Change	Initial State	Final State	Type of Heat	Example
Melting (Fusion)	Solid	Liquid	Latent Heat of Fusion	Ice melting into water
Freezing (Solidification)	Liquid	Solid	Latent Heat of Fusion	Water freezing into ice
Vaporization (Boiling/Evaporation)	Liquid	Gas	Latent Heat of Vaporization	Water boiling into steam
Condensation	Gas	Liquid	Latent Heat of Vaporization	Steam condensing into water
Sublimation	Solid	Gas	Latent Heat of Sublimation	Dry ice turning into gas
Deposition	Gas	Solid	Latent Heat of Sublimation	Frost forming from vapor

Heating and Cooling Curves



- The amount of heat energy required to change the phase of a unit mass of a substance without a change in temperature.

SI Unit: joules per kilogram (J/kg)

Types:

Latent Heat of Fusion (L_f): Heat required to convert 1 kg of a solid to a liquid at its melting point without temperature change.

Latent Heat of Vaporization (L_v): Heat required to convert 1 kg of a liquid to a gas at its boiling point without temperature change.

• Formula: $Q = m \times L$

where: L = Specific latent heat (J/kg)

Specific Latent Heat of Fusion (Water):

$$L_f = 3.34 \times 10^5 \text{ J/kg}$$

Latent Heat of Vaporization (L_v) for Water

$$L_v = 2.26 \times 10^6 \text{ J/kg}$$

- Heat is a form of energy that is transferred between objects due to a difference in temperature.
 - It flows from a hotter object to a cooler one until thermal equilibrium is reached.
- Units of Heat: joule (J) - SI unit of heat.
Calorie (cal) - Common unit in everyday use.
Conversion: $1 \text{ cal} = 4.184 \text{ J}$

1. About heat

- Temperature measures the degree of hotness or coldness of an object, indicating the average kinetic energy of its particles.

Units of Temperature: Celsius ($^{\circ}\text{C}$) - Commonly used in daily life.

Kelvin (K) - SI unit of temperature.

Conversion: $\text{K} = ^{\circ}\text{C} + 273.15$

2. Temperature

- The amount of heat energy (Q) required to raise the temperature of an object by 1°C or 1 K .

$$C' = \frac{Q}{\Delta T}$$

Where: C' = Heat capacity, Q = Heat energy (joules),

ΔT = Change in temperature ($^{\circ}\text{C}$ or K)

SI Unit: joule per kelvin (J/K)

3. Thermal (Heat) Capacity

- The amount of heat energy required to raise the temperature of 1 kg of a substance by 1°C or 1 K .

$$C = \frac{Q}{m\Delta T}$$

Where: C = Specific heat capacity, Q = Heat energy (joules),

m = Mass of the substance (kg),

ΔT = Change in temperature ($^{\circ}\text{C}$ or K)

SI Unit: Joule per kilogram per Kelvin (J/kg-K)

Relation Between Heat Capacity and Specific Heat Capacity

$$C' = m \times C$$

Where: m = Mass of the substance

4. Specific Heat Capacity

- Heat lost by hot object = Heat gained by cold object

$$m_1 c_1 (T_1 - T_f) = m_2 c_2 (T_f - T_2)$$

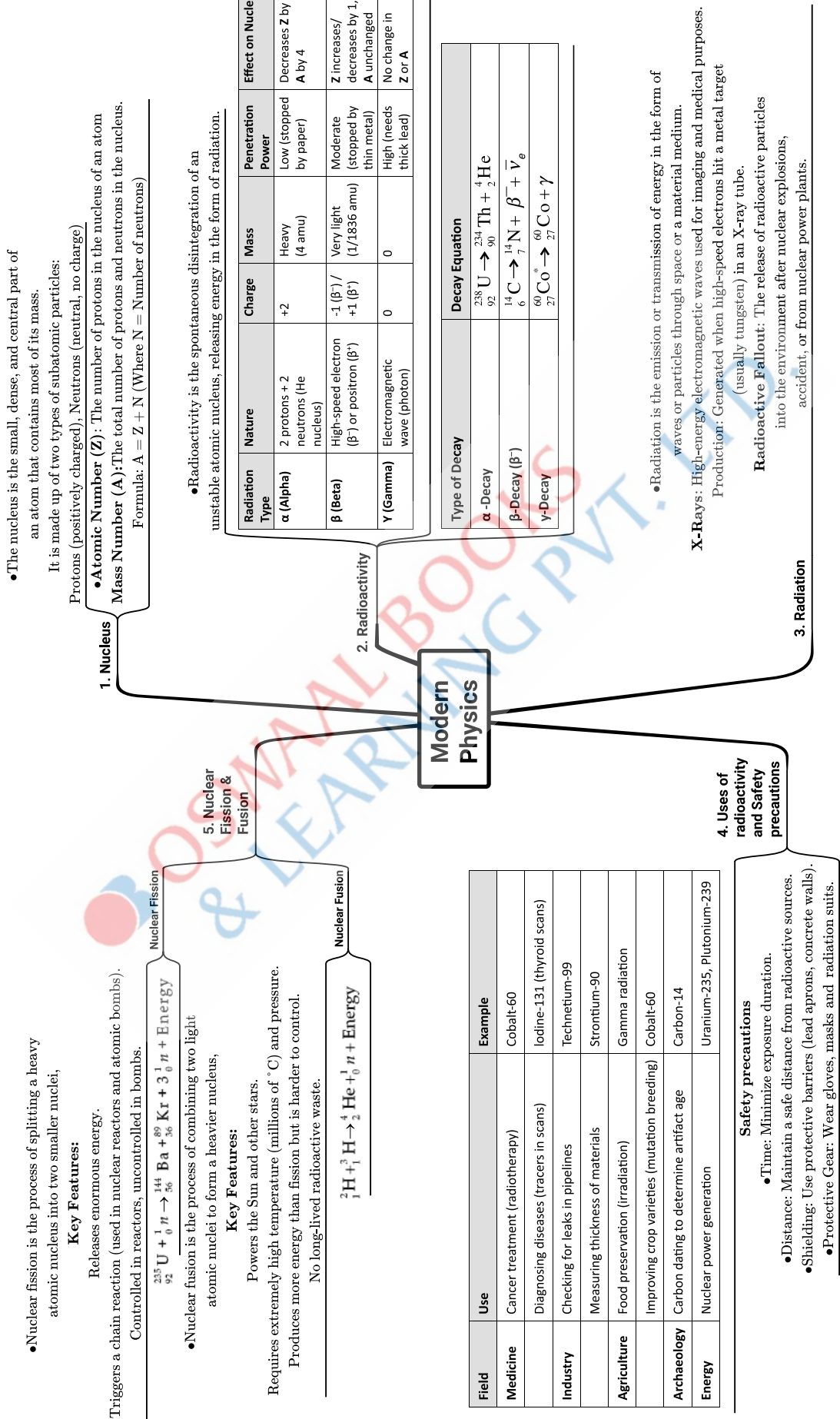
Where, m_1, m_2 = masses of hot and cold substances,

c_1, c_2 = specific heat capacities of the substances

T_1, T_2 = initial temperatures, T_f = final equilibrium temperature

5. Principle of Method of Mixtures

Heat





FORCE

TOPIC-1: TURNING FORCES, MOMENT OF FORCE (TORQUE), FORCES IN EQUILIBRIUM

Concepts covered: • Turning forces, Moment of force (Torque), Forces in equilibrium.



Revision Notes

Motion

- When a body changes its position with respect to time it is considered to be in motion.
- Types: translational, rotational, periodic and non-periodic.

Translational Motion

- It is a motion in which all the particles of an object move in a single direction and with the same velocity.
- Examples are a bus moving on a road, motion of a rectangular block down on an inclined plane.

Rotational Motion

- When a rigid body rotates about its centre of mass, it is called rotational motion. An object spinning about a fixed axis is said to be in rotational motion.
- Examples of rotational motion include a spinning top, motion of a ceiling fan, motion of a potter's wheel.

Moment of Force (Torque)

- When forces change the direction of motion and provide rotational effect in a body, then they are called **turning forces**. Turning effect of a force acting on a body

about an axis is called the **moment of the force or torque**. Torque is the measure of rotational tendency of a force.

- Torque is the product of the magnitude of force and the perpendicular distance between the line of action of force and the axis or point of rotation.
- Mathematically,

$\text{Torque} = \text{Force} \times \text{perpendicular distance from the axis of rotation.}$

- Torque is a vector quantity. Its S.I. unit is newton metre (Nm).

[Board 2023, 24]

- **Couple:** Two equal and opposite forces acting along parallel lines at different points of the body form a couple. The line along which the force acts is called the **line of action**.

Principle of Moments

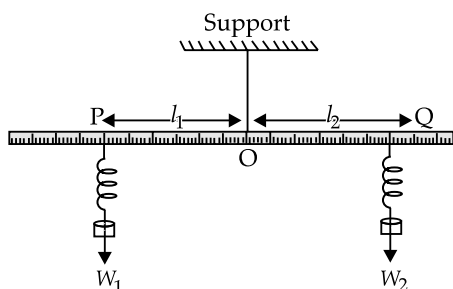
- **Principle of Moments:** When an object is in equilibrium, then the sum of the anti clockwise moments about a turning point must be equal to the sum of the clockwise moments.

Principle of Moments



Scan Me!

Force



Let the distance of the weight W_1 from the support be l_1 and the distance of the weight W_2 from the support be l_2 . The weight W_1 tries to rotate the scale in anti clockwise direction. Then,

Anti clockwise moment = $W_1 \times l_1$

And the weight W_2 tries to rotate the scale in clockwise direction. Then,

Clockwise moment = $W_2 \times l_2$

As the scale is in equilibrium, so

Total anti clockwise moment = Total clockwise moment

$$W_1 \times l_1 = W_2 \times l_2$$

- **Positive moment:** When a force acts and rotates in an anticlockwise direction, the moment is said to be positive.
- **Negative moment:** When a force rotates in a clockwise direction, the moment is considered to be negative.

Forces in Equilibrium

- When a number of forces acting on a body produce no change in its state of rest or of motion, the body is said to be in equilibrium.
- The condition for a body to be in translational equilibrium is that net force acting on the body is zero.
- The condition for a body to be in rotational equilibrium is that net torque acting on the body is zero.

MNEMONICS

Concept: Moment of force

Mnemonics: Music was Played with a Flute and Piano.

Interpretations:

M: Moment of force

P: Product of

F: Force

P: perpendicular

Moment of force = Force $\times$ perpendicular distance

Example 1

A body of mass 1.50 kg is dropped from the 2nd floor of a building which is at a height of 12 m. What is the force acting on it during its fall? ($g = 9.80 \text{ m/s}^2$)

Ans. Force acting during its free fall is given by

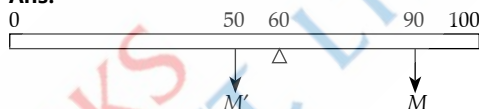
$$F = mg \\ = 1.50 \times 9.8 \text{ N} = 14.7 \text{ N}$$

And this force is independent of the height of fall.

Example 2

A uniform metre scale is kept in equilibrium when supported at the 60 cm mark and a mass M is suspended from the 90 cm mark as shown in the figure. State with reasons, whether the weight of the scale is greater than, less than or equal to the weight of mass M .

Ans.



Let the mass of the scale = M'

Applying the principle of moments,

Anti clockwise moment = clockwise moment

$$\Rightarrow M' \times (60 - 50) = M \times (90 - 60)$$

$$\Rightarrow M' \times 10 = M \times 30$$

$$\Rightarrow M' = 3M$$

Hence, the mass of the scale is more than the mass suspended.

Example 3

A force ' F ' acts on a body such that the line of action of the force passes through the point of rotation of the body. Find the magnitude of moment of the force. [A]

Ans. Here, force = F

Perpendicular distance of the force from the point of rotation = 0

$$\text{Moment of force} = F \times 0 = 0$$



Key Formula

- Principle of Moments:

Anti clockwise moment = Clockwise moment

$$W_1 l_1 = W_2 l_2$$

- Torque = Force $\times$

perpendicular distances from the axis of rotation

$$\vec{\tau} = \vec{F} \times \vec{r}$$

TOPIC-2: CENTRE OF GRAVITY, UNIFORM CIRCULAR MOTION

Concepts Covered: Centre of gravity, Uniform circular motion



Revision Notes

Centre of gravity:

➤ Centre of gravity of a rigid body is a point at which the entire weight of the body acts and algebraic sum of moments of weights of particles constituting the body is zero about this point. Its position depends on the distribution of mass. It may be within the body or outside the body. Centre of gravity and line of action of resultant force must coincide for the equilibrium. [Board 2023, 24]

➤ Centre of gravity of an irregular lamina is found by taking three points at the edges of the lamina. Then, the object is suspended from any of the chosen points, and a weighted string is dropped from the same point and a line



is drawn on the lamina along the string. This procedure is repeated from other two points on the lamina. These three lines will intersect at a point on the lamina which is basically the centre of gravity. [Board 2023, 24]

Position of centre of gravity of regular bodies:

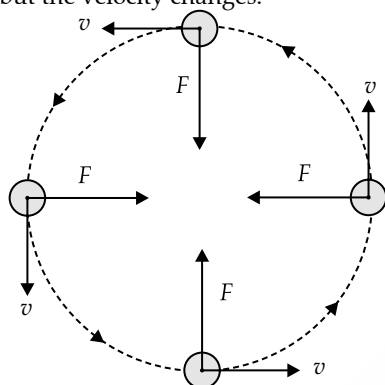
	Shape of rigid body	Position of centre of gravity
1.	Sphere	Centre of sphere
2.	Circular Disc	Centre of disc
3.	Solid / Hollow Cylinder	Middle point of the axis of the cylinder
4.	Plane Square of rectangular lamina	Point of intersection of diagonals
5.	Plane Triangular lamina	Point of intersection of medians

Force

6.	Cube	Point of intersection of diagonals
7.	Uniform rod	Centre of rod
8.	Solid Cone	On the axis at one-fourth height of the cone
9.	Hollow Cone	On the axis at one-third height of the cone

Uniform circular motion:

➤ When a body moves in a circular path around a fixed point, it is called circular motion. If the body moves with a constant speed along a circular path, it is called uniform circular motion. For example, motion of artificial satellites around the Earth. Here, the speed is constant, but the velocity changes.



➤ Due to change in velocity, there is an acceleration which is directed towards the centre of circle called



centripetal acceleration. The centripetal acceleration is given by $a = \frac{v^2}{R}$, where R is the radius of the circular orbit. If the mass of a particle is m , then according to the second law of motion $F = ma = \frac{mv^2}{R}$, and this force is

called centripetal force. It is the force required to move a body in the circular path.

➤ In circular motion, a false or pseudo force acts along the radius but directed away from the centre of the circle which is called centrifugal force. A centrifugal force formula is same as centripetal force formula only reverses the direction of the centripetal force.

Example 4

Give some examples of uniform circular motion.

Ans.(a) Motion of artificial satellites around the Earth.

(b) Movement of blades of a ceiling fan.

(c) Revolution of the Earth around the Sun.



Key Terms

➤ Centripetal acceleration $a = \frac{v^2}{R}$, where R is the radius of the circular orbit.

➤ Centripetal force $F = ma = \frac{mv^2}{R}$



WORK, POWER AND ENERGY

TOPIC-1: WORK, ENERGY, DIFFERENT TYPES OF ENERGY

Concepts Covered: Work, Energy, Different types of energy.



Revision Notes

Work

- Work is said to be done only when force applied on a body makes the body move. S.I. unit of work is joule (J).
- Work is equal to the product of force (F) and displacement (s) in the direction of force.

$$W = F \times s$$

[Board 2023, 24]

- When a non-zero force (F) is applied on an object, it displaces by (s) by making an angle (θ) with the direction of force. Then work done mathematically, is defined as:

$$W = Fs \cos \theta$$

- Work can be positive, negative or zero.
- If the angle between force and the displacement is acute, then work is said to be **positive**.
- If the angle between force and the displacement is 90° i.e., displacement is perpendicular to the force applied, work is said to be **zero**.
- If the angle between force and the displacement is obtuse, the work is said to be **negative**.

[Board 2022]

- Work done can be zero if :

- force applied is zero i.e., no force acts on the body.
- displacement of body is zero.
- angle between force and displacement is 90° .

- When a coolie walks horizontally while carrying a load on his head, no work is done against the force of gravity.
- When a body rotates in a circular path, no work is done against the centripetal force, as force and displacement are normal to each other.

- C.G.S. unit of energy or work is erg.

$$1 \text{ joule} = 10^7 \text{ erg}$$

- One joule of work is said to be done when a force of 1 newton displaces a body through a distance of 1 metre in its own direction.

- 1 erg of work is said to be done when a force of 1 dyne displaces a body through a distance 1 cm in its own direction.

Energy

- Energy is the capacity of a body to do work. Its S.I. unit is joule.

- One kilowatt hour (1 kWh) is the energy spent (or work done) by a source of power 1 kW in 1 hour.

- One electronvolt is the measure of an amount of kinetic energy gained by an electron accelerating through potential difference of one volt.

Work and Energy



Scan Me!

Different types of energy

➤ Energy exists in different forms like mechanical energy, chemical energy, nuclear energy, sound energy, light energy.

➤ Mechanical energy is of two types i.e.,

(i) Kinetic energy = $\frac{1}{2}mv^2$

(ii) Potential energy = mgh

[Board 2022]

➤ Kinetic energy is the energy possessed by a body by virtue of its motion. Examples include a moving train, a running boy, etc.

➤ **Types of kinetic energy :**

(i) Translation kinetic energy (e.g., a car moving in straight path, a freely falling body possesses translational kinetic energy)

(ii) Rotational kinetic energy (e.g., a spinning top, a rotating fan possesses rotational kinetic energy)

(iii) Vibrational kinetic energy (e.g., a wire clamped at both the ends when struck in the middle vibrates, possessing vibrational kinetic energy)

➤ **Derivation of expression of Kinetic energy, $K = \frac{1}{2}mv^2$**

A body of mass m , initially at rest, is acted upon by a constant force ' F ' which produces acceleration and the body gains a final velocity v .

Force, $F = \text{mass} \times \text{acceleration} \quad \dots(i)$

Using 3rd kinematic equation of motion, $v^2 = u^2 + 2as \quad \dots(ii)$

Where, initial velocity, $u = 0$

Final velocity, v

Acceleration, a

So, the equation (ii) becomes

$$v^2 = \frac{v^2}{2a} \quad \dots(iii)$$

Since, change in Kinetic energy is equal to the amount of work done by the body.

Change in Kinetic energy = $F \times s$

$$K_f - K_i = ma \times \frac{v^2}{2a}$$

since, $u = 0$; $K_i = 0$, $K_f = K$ (suppose) [Using equation (i) and (iii)]

$$K = \frac{1}{2}mv^2$$

➤ **Relation of Kinetic energy with momentum**

Momentum = $P = mv$

Kinetic energy $K = \frac{1}{2}mv^2$

Or, $K = \frac{m^2v^2}{2m}$

Or, $K = \frac{p^2}{2m}$

$$\therefore p^2 = 2mK$$

➤ Potential energy is the energy possessed by a body by virtue of its position or configuration.

➤ **Potential energy is of two types:**

(i) Elastic potential energy

(ii) Gravitational potential energy

➤ **Derivation of expression of potential energy, $U = mgh$**

Let a body of mass ' m ' be lifted upwards to a height ' h ' above the ground.

Then, work done on the body against the force of gravity = force $\times$ displacement

Force, $F = mg$

Work done, $W = mg \times h$

This work done is stored in the body in form of potential energy, $U = mgh$

Example 1

A body of mass 0.2 kg falls from a height of 10 m to a height of 6 m above the ground. Find the loss in potential energy taking place in the body. ($g = 10 \text{ ms}^{-2}$)

Ans. Given: Mass, $m = 0.2 \text{ kg}$

Initial height, $h_1 = 10 \text{ m}$

final height, $h_2 = 6 \text{ m}$

P.E.₁ = $0.2 \times 10 \times 10 = 20 \text{ J}$

P.E.₂ = $0.2 \times 10 \times 6 = 12 \text{ J}$

$\therefore$ Loss in potential energy = $(20 - 12) \text{ J} = 8 \text{ J}$

Example 2

A moving body weighing 400 N possesses 500 J of kinetic energy. Calculate the velocity with which the body is moving. ($g = 10 \text{ m/s}^2$)

Ans. Weight of body = 400 N

Let mass of body = $m \text{ kg}$

$$\Rightarrow mg = 400$$

$$\Rightarrow m \times 10 = 400$$

$$\Rightarrow m = 40 \text{ kg}$$

Given, kinetic energy = 500 J

Let its velocity be $v \text{ m/s}$

$$\therefore \frac{1}{2}mv^2 = 500 \text{ J}$$

$$\Rightarrow \frac{1}{2} \times 40 \times v^2 = 500$$

$$\Rightarrow v^2 = \frac{500}{20}$$

$$\Rightarrow v^2 = 25$$

$$\Rightarrow v = 5 \text{ ms}^{-1}$$

MNEMONICS

Concept: Positive, Negative and Zero work done.

Mnemonics: Appu Planned On New Year Day to visit New Zealand.

Interpretations:

A: Acute

P: Positive

O: Obtuse

N: Negative

N: Ninety

Z: Zero

When angle between the force and displacement is acute, the work done is positive.

When angle between the force and displacement is obtuse, the work done is negative.

When angle between the force and displacement is ninety degree, the work done is zero.



Key Equations

- Work = Force $\times$ displacement (in the direction of force)
- Work, $W = F \cos \theta$
- Gravitational potential energy, $U = mgh$
- Kinetic energy, $K = \frac{1}{2} mv^2$
- 1 joule = 10^7 ergs
- 1 eV = 1.6×10^{-19} J
- $p^2 = 2 mK$

Where, p = Linear momentum

and K = Kinetic energy

TOPIC-2: POWER AND TRANSFORMATION OF ENERGY

Concepts Covered: Power, Sources of energy, Conversion of energy from one form to another



Revision Notes

Power

➤ Power is the rate of doing work. Its S.I. unit is watt.

➤ $1 \text{ watt} = 1 \text{ Js}^{-1}$

➤ If one joule of work is done in 1 second, the power spent is said to be 1 watt.

➤ $1 \text{ horse power (hp)} = 746 \text{ Watts} = 0.746 \text{ kW}$

➤ $1 \text{ kW} = 1000 \text{ W}$, $1 \text{ MW} = 10^6 \text{ W}$ and $1 \text{ GW} = 10^9 \text{ W}$

➤ The C.G.S unit of power is erg per second.

➤ $1 \text{ watt} = 1 \text{ J s}^{-1} = 10^7 \text{ erg s}^{-1}$

Transformation of Energy

Energy exists in different forms. The change of one form of energy into another form of energy is called transformation of energy.

[Board 2024]

➤ Solar energy is the energy radiated by the Sun.

➤ Solar panels, solar furnaces and solar cells use solar energy to do useful work.

➤ The energy released on burning coal, oil, wood or gas is the heat energy.

➤ Light is also a form of energy. We can see objects in presence of light only.

➤ The energy contained in fossil fuels such as coal, petroleum and natural gas is called chemical energy.

➤ The energy possessed by the fast moving water is called the hydro energy. It is used to generate electricity.

➤ The energy released due to loss in mass during nuclear reaction is called nuclear energy.

➤ The heat energy stored in the core of Earth is called geothermal energy.

➤ The energy possessed by the fast-moving air is called wind energy. Windmills use this energy to produce electricity.

➤ Sound energy is possessed by vibrating bodies.

[Board 2024]



Example 3

A crane 'A' lifts a heavy load in 5 seconds, whereas another crane 'B' does the same work in 2 seconds. Compare the power of crane 'A' to that of crane 'B'

[ICSE 2020]

Ans. Crane A lifts a heavy load in 5 second and crane B does the same work in 2 second.

Power = Work done / Time taken

For crane A, $P_A = \frac{W}{5}$

For crane B, $P_B = \frac{W}{2}$

$\therefore \text{Power of crane A : Power of crane B} = \frac{\frac{W}{5}}{\frac{W}{2}} = 2 : 5$



Key Equations

➤ Power, $P = \frac{\text{Work done}}{\text{Time taken}} = \frac{W}{t}$

➤ Power, $P = \frac{\text{Energy used}}{\text{Time taken}} = \frac{E}{t}$

➤ Power = Force $\times$ Velocity = $F \times v$

➤ $1 \text{ watt hour} = 3600 \text{ J} = 3.6 \text{ kJ}$

➤ $1 \text{ kilowatt hour} = 3.6 \times 10^6 \text{ J}$

➤ $1 \text{ calorie} = 4.186 \text{ joule}$

MNEMONICS

Concept: Energy Sources

Mnemonics: Champak arrived Nasik via Rajkot.

Nitin Chopra received him.

Interpretations:

C: Conventional Energy Sources

N: Non

R: Renewable

N: Non

C: Conventional Energy Sources

R: Renewable

3

Machines

TOPIC-1: SIMPLE MACHINES

Concepts Covered: Machines as Force Multipliers, Mechanical Advantage, Velocity Ratio, Efficiency



Revision Notes

- A **machine** is a device by which we can either overcome a large resistive force (or load) at some point by applying a small force (or effort) at a convenient point and in a desired direction or by which we can obtain a **gain** in speed.
- The external **resistance** against which any machine acts is called **load (L)**.
- The force applied on the machine to overcome the load is called **effort (E)**.

- The ratio of the load to the effort is called the mechanical advantage of the machine, i.e.,

$$\text{Mechanical advantage (M.A.)} = \frac{\text{Load (L)}}{\text{Effort (E)}} \quad [\text{Board 2024}]$$

- Mechanical advantage is a ratio. Hence, it has no unit.
- The ratio of the velocity of effort to the velocity of load is called the velocity ratio of machine i.e.,

$$\text{Velocity ratio (V.R.)} = \frac{\text{Velocity of effort (V}_E\text{)}}{\text{Velocity of load (V}_L\text{)}} \quad [\text{Board 2024}]$$

- The velocity ratio is also defined as the ratio of the displacement of effort to the displacement of load.



- Velocity ratio is a ratio. Hence, it has no unit.
- A machine in which there is no loss of energy is called an ideal machine.
- The efficiency of an ideal machine is 100% but in practice, it is not possible.
- No machine does work by itself. For an ideal machine, output = input.
- A machine is a device that makes work easier for us. It does so by enabling us to:
 - multiply force
 - apply force at a convenient point or in a convenient direction and
 - obtain gain in speed
- The efficiency of a machine is the ratio of the useful work done by the machine (the output) to the total work done on the machine (input)

$$\text{Efficiency, } \eta = \frac{\text{output}}{\text{input}} \times 100\% \quad [\text{Board 2024}]$$

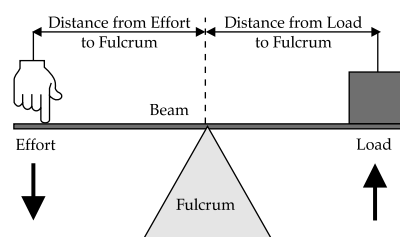
- For a practical machine,

$$\text{Efficiency} = \frac{\text{Mechanical advantage}}{\text{Velocity ratio}}$$



Machines

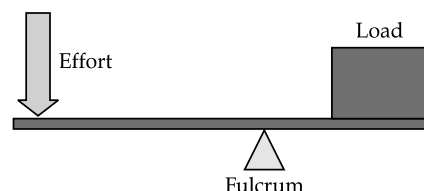
➤ A lever is a rigid beam that rotates around one point to move a load by applying a force to a third point.



➤ **Law of the Lever:** $\text{Load} \times \text{Load arm} = \text{Effort} \times \text{Effort arm}$

➤ **Types of Lever:**

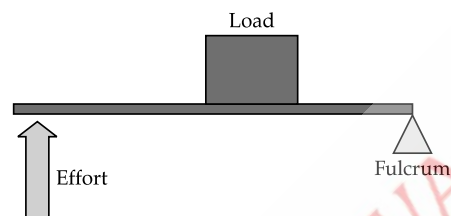
• **First class:** Fulcrum is located between the input effort and the output load.



Mechanical advantage may be greater than, less than or equal to 1 depending on the position of fulcrum.

Examples: crowbar, scissors, pliers, seesaw etc.

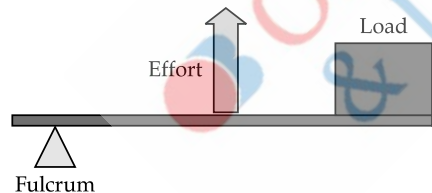
• **Second class:** Load is between fulcrum and Effort.



Mechanical advantage is always greater than 1.

Examples: nut cracker, wheel barrow etc.

• **Third class:** Effort is between fulcrum and Load.



Mechanical advantage is always less than 1.

Examples: tweezers, fishing Rod, human arm etc.

Example 1

A lever is used to lift a 500 N load. The effort arm is 2 meter long, and the load arm is 0.5 meter long. Calculate the effort required to lift the load.

Ans. • Principle of Lever:

$$\text{Effort} \times \text{Effort Arm} = \text{Load} \times \text{Load Arm}$$

• **Given:**

- o Load (L) = 500 N
- o Load arm = 0.5 m
- o Effort arm = 2 m

• **To find:** Effort (E)

• **Formula:** $E \times \text{Effort arm} = L \times \text{Load arm}$

• **Substitute values:** $E \times 2 \text{ m} = 500 \text{ N} \times 0.5 \text{ m}$

• $E = (500 \text{ N} \times 0.5 \text{ m}) / 2 \text{ m}$

• **Effort:** $E = 125 \text{ N}$

Therefore, an effort of 125 N is required to lift the 500 N load.



Key Formula

➤ **Mechanical Advantage (M.A)**

$$\text{MA} = \frac{\text{Load}}{\text{Effort}}$$

Where,

Load: The weight or force being overcome by the machine.

Effort: The force applied to the machine to operate it.

➤ **Velocity Ratio (V.R)**

$$\text{V.R} = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$$

This represents the ideal mechanical advantage, assuming no friction.

➤ **Efficiency of a Machine**

$$\text{Efficiency} = \left(\frac{\text{Mechanical Advantage}}{\text{Velocity Ratio}} \right) \times 100\%$$

Efficiency is always below 100% because of energy losses caused by friction.

➤ **Law of Conservation of Energy**

$$\text{Input Energy} = \text{Useful Output Energy} + \text{Wasted Energy}$$

This principle applies to all machines, highlighting that energy cannot be created or destroyed, only transformed.

Specific Formulas for Simple Machines:

• **Lever:** $\text{Effort} \times \text{Effort Arm} = \text{Load} \times \text{Load Arm}$

TOPIC-2: PULLEY SYSTEMS

Concepts Covered: Pulley, Block and Tackle



Revision Notes

- A **pulley** is simply a grooved wheel that is used along with a rope or a chain.
- A pulley which has its axis of rotation fixed in position is called a **fixed pulley**.
- For a pulley, mechanical advantage = $\frac{\text{Load}}{\text{Effort}}$ [Board 2022, 23, 24]
- For a **single fixed pulley**, ideal M.A. = 1 and V.R. = 1.
- A pulley whose axis of rotation is movable is called a movable pulley. A single movable pulley can act as a force multiplier.
- For a **single movable pulley**, ideal M.A. = 2 and V.R. = 2.
- We use single fixed pulley to change the direction of application of effort.
- A combination of pulleys enables us to multiply force by a factor that is dependent on the number of strands used to support the load.
- The block and tackle system of pulleys is made by having two blocks of pulleys in which the lower block is movable but the upper one is attached to a fixed support.
- When the weight of the lower movable block of pulley is negligible as compared to that of load and there is no friction, then M.A. = V.R. = n = number of strands of tackle supporting the load.

Comparison between single fixed pulley and single movable pulley



Scan Me!

Pulley



Scan Me!

Example 1

A woman draws water from a well using a fixed pulley. The mass of the bucket and water together is 6 kg. The force applied by the woman is 70 N.

Calculate the mechanical advantage.

(take $g = 10 \text{ ms}^{-2}$)

Ans. Given,

Load, $L = mg = 6 \times 10 = 60 \text{ N}$

Effort, $E = 70 \text{ N}$

Mechanical Advantage M.A. = $\frac{\text{Load}}{\text{Effort}}$

$$\text{M.A.} = \frac{60}{70} = 0.857$$

Example 2

A single fixed pulley is used to lift a 20 kg object. Calculate the effort required, assuming no friction.

Ans. • **Principle:** In an ideal single fixed pulley,

Effort = Load

• **Given:**

Load (L) = 20 kg

• **To find:** Effort (E)

• **Formula:** $E = L$

• **Substitute values:** $E = 20 \text{ kg}$

Therefore, an effort of 20 kg is required to lift the 20 kg object.

MNEMONICS

Concept: M.A. and V.R. of single fixed and movable pulley.

Mnemonics : Farhad stood 1st and Madhavan stood 2nd.

Interpretation :

↓	↓	↓	↓
Fixed Pulley (single)	1 (Both M.A. and V.R.)	Movable Pulley (single)	2 (Both M.A. and V.R.)

**Key Equation**

- Mechanical advantage = $\frac{\text{Load}}{\text{Effort}}$

For single fixed pulley, M.A. = 1, V.R. = 1

For single movable pulley, M.A. = 2, V.R. = 2

For block and tackle system, M.A. = V.R. = n

(n = number of strands of tackle system supporting the load)

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Light

TOPIC-1: REFRACTION OF LIGHT AT PLANE SURFACE AND TOTAL INTERNAL REFLECTION

Concepts Covered: Refraction of light through a glass block and a triangular prism, total internal reflection



Revision Notes

- The speed of light in air/vacuum is 3×10^8 m/s.
- A medium is said to be optically denser if light slows down in it.
- A medium is said to be rarer if light speeds up in it.
- When a ray of light travels from a rarer medium to a denser medium, it bends towards the normal.
- When a ray of light travels from a denser medium to a rarer medium, it bends away from the normal.
- The conditions when light travelling from one medium to another goes undeviated.
 - (i) Optical densities of both the media are the same.
 - (ii) Angle of incidence is zero, i.e., light falls normally on the surface.
- Refractive index has no unit. **[Board 2022, 24]**
- When light passes from one medium to another, its frequency does not change but wavelength, speed and direction changes.
- When light passes from rarer to denser medium, its wavelength decreases.
- When light passes from denser medium to rarer medium, its wavelength increases.
- In case of minimum deviation of light while passing through the prism, the refracted ray inside the prism is parallel to the base of the prism.
- **Factors affecting the angle of deviation of light travelling through the prism are:**
 - (i) the angle of incidence.
 - (ii) the material of the prism (i.e., refractive index).
 - (iii) the angle of prism (A).
 - (iv) the colour or wavelength (λ) of light used.

[Board, 2024]

➤ **Factors affecting lateral displacement of light passing through a rectangular glass block:**

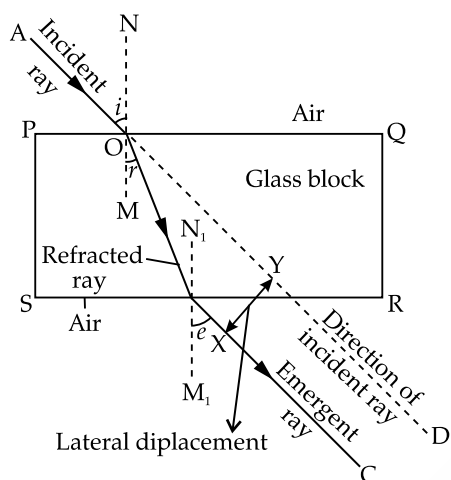
- (i) The thickness of glass block
- (ii) The angle of incidence
- (iii) The refractive index of the glass
- (iv) The wavelength of light used
- Cause of refraction is that light has different speeds in different media.
- The refractive index of a transparent medium is always greater than 1.

Light

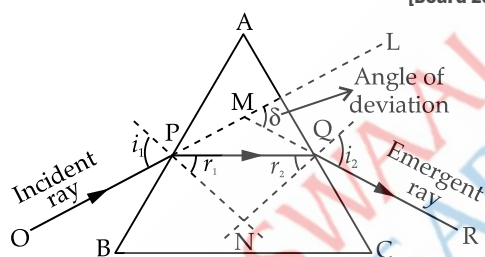
➤ Factors affecting refractive index of a medium:

- (i) Nature of medium
- (ii) Physical conditions such as temperature
- (iii) The colour or wavelength of light [Board, 2024]
- Speed of light in glass is 2×10^8 m/s and in water is 2.25×10^8 m/s.
- Refractive index of glass is 1.5, of water is 1.33 and of diamond is 2.41.

➤ Refraction of light through a rectangular glass block [Board, 2022]

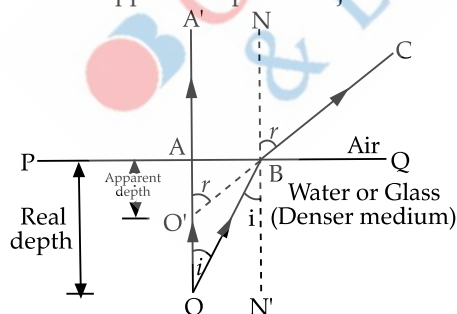


➤ Refraction of light through a glass prism [Board 2022, 23]

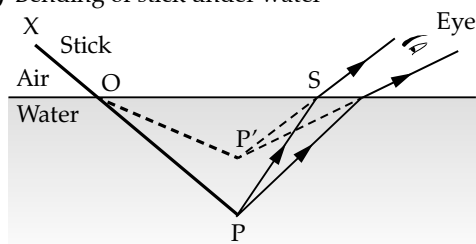


➤ Applications of refraction of light:

(a) Real and apparent depths of object in water



(b) Bending of stick under water

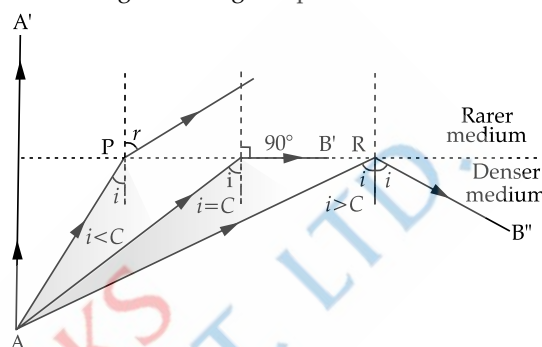


➤ Critical angle is the angle of incidence in the denser medium for which angle of refraction in the rarer medium is 90° . [Board, 2023]

➤ When a ray of light travels from a denser to a rarer medium with an angle of incidence greater than critical angle, then no refraction takes place and the entire light is reflected back in the denser medium. This is known as total internal reflection.

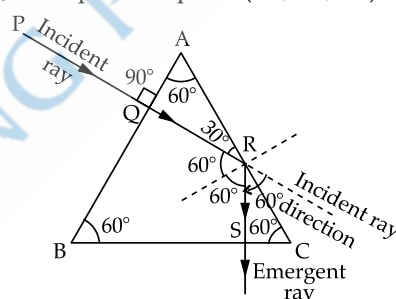
➤ Essential conditions for total internal reflection to take place:

- (i) Light must travel from a denser to a rarer medium.
- (ii) The angle of incidence should be greater than the critical angle for the given pair of media.

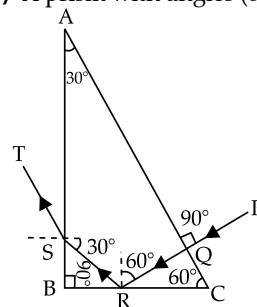


➤ Total internal reflection in a triangular glass prism: [Board 2024, 23]

(a) An equilateral prism ($60^\circ, 60^\circ, 60^\circ$)



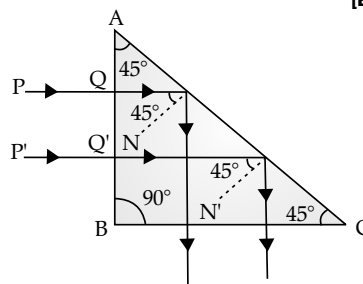
(b) A prism with angles ($60^\circ, 30^\circ, 90^\circ$) [Board 2024, 23]



(c) A prism with angles ($45^\circ, 45^\circ, 90^\circ$)

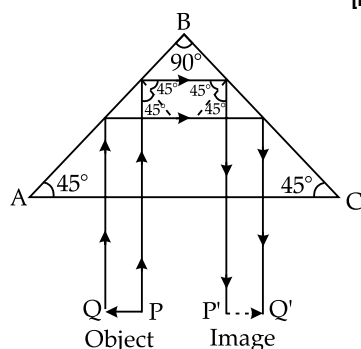
(i) To deviate the incident ray by 90° , i.e., ($\delta = 90^\circ$)

[Board 2024, 23]



- (ii) To deviate the incident ray by 180° , i.e., ($\delta = 180^\circ$)

[Board 2024,23]



➤ **Comparison between total internal reflection and reflection from a plane mirror:**

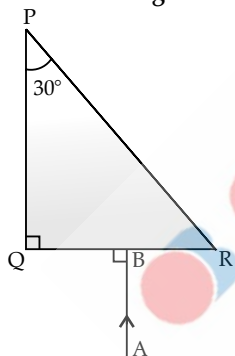
(i) In total internal reflection, light enters from a denser to a rarer medium, whereas in reflection from a plane mirror, light can be incident from any medium.

(ii) There is no loss of energy in total internal reflection, as the entire light is internally reflected, whereas in case of reflection from plane mirror, there is a loss of energy due to absorption and refraction of light.

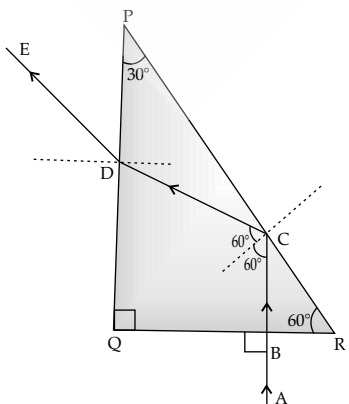


Example 1

Complete the path of the ray AB through the glass prism in PQR, till it emerges out of the prism. Given the critical angle of the glass as 42° .



Ans.



Key Terms

➤ **Refraction:** The change in direction of the path of light, when it passes from one transparent medium to another transparent medium is called refraction. It is a surface phenomenon.

➤ **Denser Medium:** A medium is said to be optically denser if the speed of light in it decreases.

➤ **Rarer Medium:** A medium is said to be optically rarer if the speed of light in it increases.

➤ **Angle of Incidence:** It is the angle between incident ray and the normal to the surface at point of incidence.

➤ **Angle of Refraction:** It is the angle between the refracted ray and the normal at the point of incidence.

➤ **Laws of Refraction:**

➤ **First Law:** The incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane.

➤ **Second Law:** The ratio of the sine of the angle of incidence i to the sine of the angle of refraction r is constant for the pair of given media and for a monochromatic light. This constant is called refractive index. This law is also called Snell's Law.

➤ **Refractive Index:** The refractive index of the second medium with respect to the first medium is defined as the ratio of the sine of the angle of incidence in the first medium to the sine of the angle of refraction in the second medium.

➤ **Lateral Displacement:** The distance between the incident ray extrapolated and the emergent ray when light travels through a rectangular glass slab is called lateral displacement.

➤ **Angle of Deviation:** The angle between the emergent ray and the incident ray extrapolated when light passes through a prism is called angle of deviation.

➤ Frequency, $f = \frac{\text{Speed of light in medium, } v}{\text{Wavelength of light in that medium, } \lambda}$

➤ Refractive index (or absolute refractive index), $\mu = \frac{\text{Speed of light in air, } c}{\text{Speed of light in that medium, } v}$

➤ Refractive index of second medium with respect to first medium = $\frac{\text{Absolute refractive index in medium 2, } \mu_2}{\text{Absolute refractive index in medium 1, } \mu_1}$

➤ $A + \delta = i + e$, where A is angle of prism, δ is the angle of deviation, i is the angle of incidence and e is the angle of emergence.

➤ $\mu = \frac{\text{Real depth}}{\text{Apparent depth}}$

➤ $\mu = \frac{1}{\sin C}$, where C is the critical angle.

MNEMONICS

Concept: Movement of light ray when refracted from rarer to denser medium

Mnemonics: Watch RD's Movie ToNight

Interpretation:

R: Rarer to

D: Denser

M: Moves

T: Towards

N: Normal

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TOPIC-2: REFRACTION THROUGH A LENS

Concepts Covered: Lenses (Convex and Concave)



Revision Notes

➤ A lens is a transparent refracting medium bounded by two surfaces, both of which are either spherical in shape or one is plane and other is spherical. A lens may be regarded as being made up of a set of prisms. (A lens is not made up of prisms.)

➤ There are mainly two types of lenses:

- (i) Convex or converging lens
- (ii) Concave or diverging lens

➤ Convex or converging lenses are thin at the edges and thick at the middle.

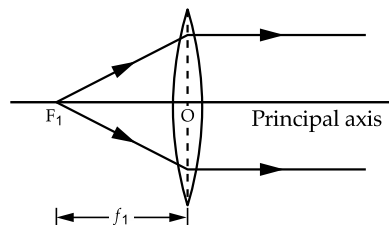
➤ Concave or diverging lenses are thick at the edges and thin at the middle.

➤ The **principal axis** of a lens is the line joining the centres of the two spheres of the two surfaces of which lens is a part.

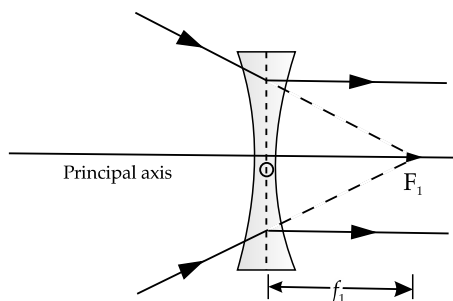
➤ **Optical centre** of a thin lens is the point on the principal axis of the lens through which a ray of light passes undeviated.



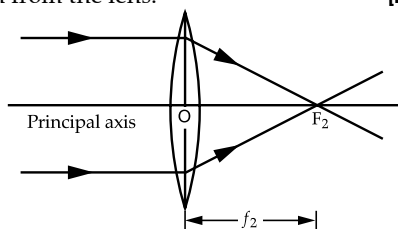
➤ For a convex lens, the first focal point is a point F_1 on the principal axis of the lens such that the rays of light coming from it become parallel to the principal axis of the lens after refraction from the lens. [Board, 2024]



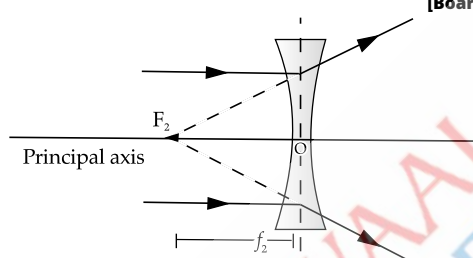
➤ For a concave lens, first focal point is a point F_1 on the principal axis of the lens such that the incident rays of light appearing to meet at it become parallel to the principal axis of the lens after refraction from the lens.



- For a convex lens, the second focal point is a point F_2 on the principal axis of the lens such that the rays of light incident parallel to the principal axis pass through it after refraction from the lens. [Board, 2024]



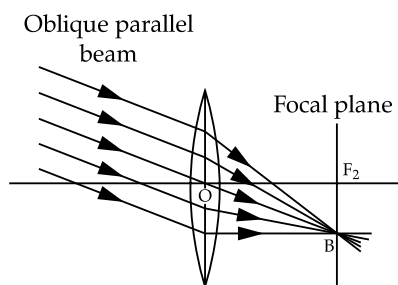
- For a concave lens, the second focal point is a point F_2 on the principal axis of the lens such that the rays of light incident parallel to the principal axis appear to be diverging from this point after refraction from the lens. [Board, 2023]



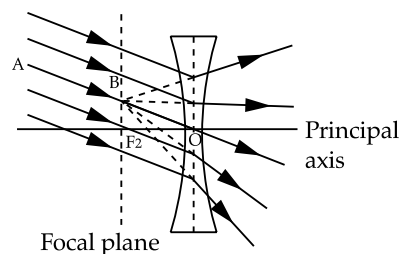
- A concave lens always produces a virtual, erect and diminished image of a real object.
- A convex lens can produce inverted real images - magnified, diminished and of same size. It can also produce magnified erect virtual images.
- The power of a lens is the reciprocal of its focal length measured in metre. It is measured in units of **diopetre (D)**.
- A convex lens of small focal length may be used as a simple magnifying glass or a reading lens. For this, the object is kept between the optical centre and the focus of the lens. When it is used in this manner, it is also known as a **simple microscope**.

- The magnification produced by a lens is the ratio of a size of the image produced by it to the size of the object.

- **Refraction of an oblique parallel beam by a convex lens:**



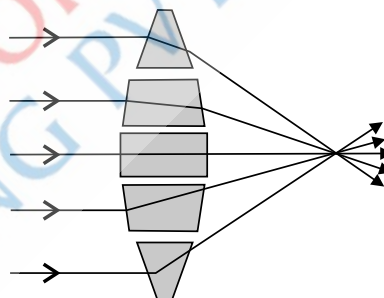
- **Refraction of an oblique parallel beam by a concave lens:**



- A convex lens is considered to be made up of a large number of truncated prisms and a glass slab.

A convex lens may be considered to be a combination of a large number of truncated prisms. The prisms in the upper half have their bases downwards and the prisms in the lower half have their bases upwards with continuously changing angle of each prism. The central part of the lens is like a glass slab.

Rays of light passing through the prisms tend to deviate towards the base. The central rectangular glass slab allows the incident ray to pass undeviated.

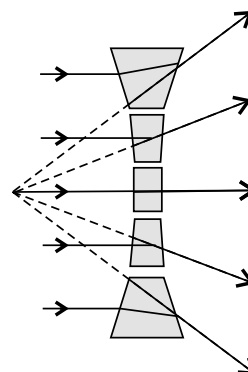


- A concave lens is considered to be made up of a large number of truncated prisms and a glass slab.

The prisms in the upper half have their bases upwards and the prism in the lower half have their bases downwards with continuously changing angle of each prism.

The central part is like a rectangular glass block.

Rays of light passing through the prisms tend to deviate away from the base. The central rectangular glass slab allows the incident ray to pass undeviated.



➤ **Distinction between a real and virtual image:**

S. No.	Real Image	Virtual Image
1.	A real image is formed due to actual intersection of the rays refracted by the lens.	A virtual image is formed when the rays refracted by the lens appear to meet if they are produced backwards.
2.	A real image can be obtained on a screen.	A virtual image cannot be obtained on a screen.
3.	A real image is inverted with respect to the object. For example, the image of a distant object formed by a convex lens.	A virtual image is erect with respect to the object. For example, the image of a distant object formed by a concave lens.

➤ **Relative positions of the object and image in a convex lens.**

Positions of Object	Position of Image	Size of Image	Name of Image	Application
1. At infinity	At F_2	Highly diminished	Real and inverted	Burning glass
2. Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted	Camera lens
3. At $2F_1$	At $2F_2$	Same size	Real and inverted	Terrestrial telescope
4. Between F_1 and $2F_1$	Beyond $2F_2$	Magnified	Real and inverted	Slide projector
5. At F_1	At infinity	Highly magnified	Real and inverted	Collimator of spectrometer
6. Between the optical centre and F_1	On same side, behind the object	Magnified	Virtual and upright	Corrective lens or magnifying glass

➤ **Relative positions of object and image in a concave lens.**

Position of the Object	Position of the Image	Nature of the Image	Size of the Image	Application
1. At infinity	At the focus, on the same side of the lens as the object	Virtual and upright	Highly diminished	Galilean telescope
2. At any position between infinity and optical centre	Between the focus and optical centre, on the same side of the lens as the object	Virtual and upright	Diminished	Corrective lens for myopic eye

➤ **Difference between the image formed by a convex lens and by a concave lens.**

S. No.	Image by a Convex Lens	Image by a Concave Lens
1.	The image may be real as well as virtual. It is real if the object lies at or beyond focus, while it is virtual if the object lies between focus and optical centre.	The image is always virtual for all positions of the object.
2.	The image may be magnified, of the same size, as well as diminished.	The image is always diminished.
3.	The image may be inverted, as well as erect. The image is inverted if the object is at or beyond focus and erect if the object is between focus and optical centre.	The image is always erect.

➤ The deviation produced by a lens in the path of rays refracted through it is a measure of its power. Power of a lens is positive for the convex lens and negative for the concave lens.

[Board, 2024]

➤ Power of lens (in diopetre) = $\frac{1}{f}$

➤ **Magnifying Power:** The magnifying power of the microscope is given as:

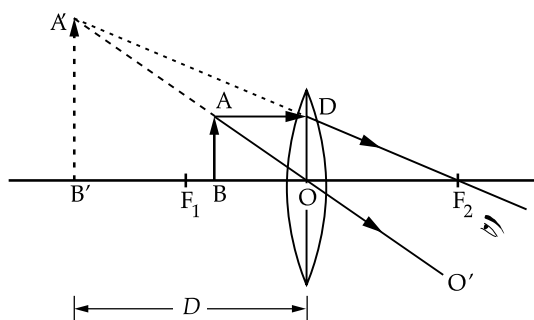
$$\text{Magnifying power} = 1 + \frac{D}{f}$$

Where, D = least distance of distinct vision

f = focal length of the lens

➤ **Ray diagram for location of image in magnifying glass:**

Light



➤ **Focal Length of Lens:** The distance between focus and optical centre of the lens is called **focal length of a lens**.

➤ **Focal Plane:** The plane passing through the focus and perpendicular to the principal axis is called **focal plane**.

➤ **Aperture:** The effective diameter of the circular outline of a spherical lens is called its **aperture**.

➤ **Centre of Curvature:** Centre of curvature of a surface of a lens is the centre of the sphere of which lens is a part. A lens having two spherical surfaces has two centres of curvature.

➤ **Radius of Curvature:** Radius of curvature is the distance between the optical centre and centre of curvature.

➤ **Sign Conventions:**

- All distances are measured from the optical centre of the lens.
- Distances measured in the direction of the incident ray are taken as positive and opposite to the direction of the incident ray are taken as negative.
- Distances measured upwards and perpendicular to principal axis are taken as positive, whereas distances measured downwards and perpendicular to principal axis are taken as negative.

➤ Lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Where, u = Object distance (always negative)

v = Image distance

(may be positive or negative)

f = Focal length (positive for convex lens and negative for the concave lens)

➤ Lenses are used for eye defect correction, magnifying glass, telescope, camera.

TOPIC-3: SPECTRUM, SCATTERING OF LIGHT

Concepts Covered: Electromagnetic spectrum, dispersion, Scattering of light



Revision Notes

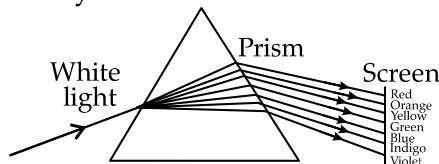
➤ The phenomenon of splitting of white light by a prism into its constituent colours is known as **dispersion**.

➤ The band of colours seen on passing white light through a prism is called the **spectrum**.

➤ **Cause of Dispersion:**

➤ The cause of dispersion is the change in speed of light with wavelength in dispersive medium. When white light enters the first surface of a prism, light of different colours due to their different speeds in the glass gets deviated toward the base of prism through different angles.

➤ **Dispersion by a Prism:**



➤ The angle of deviation depends upon (i) angle of incidence at first surface, (ii) angle of prism and (iii) refractive index of the material.



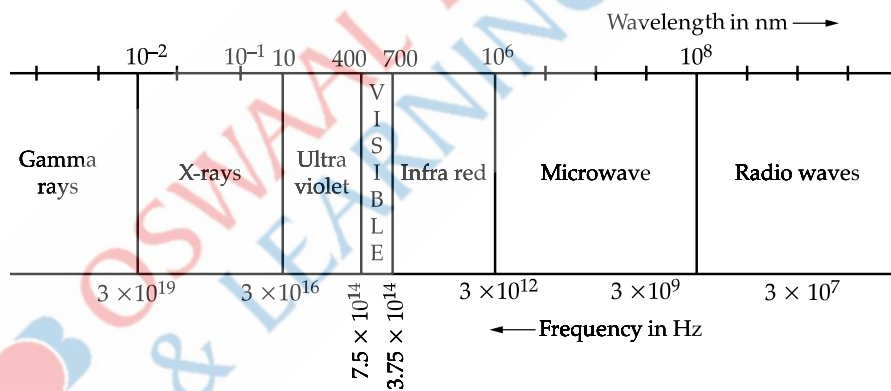
➤ **Wavelengths and frequencies of different colours in white light:**

Colour	Wavelength Range (Nearly)	Frequency Range in 10^{14} Hz
Violet	4000–4460 Å	7.5–6.73
Indigo	4460–4640 Å	6.73–6.47
Blue	4640–5000 Å	6.47–6.01
Green	5000–5780 Å	6.01–5.19
Yellow	5780–5920 Å	5.19–5.07
Orange	5920–6200 Å	5.07–4.84
Red	6200–7000 Å	4.84–3.75

➤ The complete electromagnetic spectrum in the increasing order of their wavelength (or decreasing order of their frequency) is given below: **[Board, 2024]**

(1) Gamma rays, (2) X-rays, (3) ultraviolet rays, (4) visible light, (5) infrared radiations, (6) microwaves and (7) Radio waves

Thus, infrared spectrum is the part of the spectrum just beyond the red end, while the ultraviolet spectrum is the part of the spectrum just before the violet end.



Electromagnetic Spectrum

Name of the Wave	Frequency in Hz	Discoverer	Source	Method of Detection
Gamma rays	Above 3×10^{19}	Paul Villard	In cosmic rays, in radiations from radioactive substances	By their large penetrating power
X-rays	$3 \times 10^{19} - 3 \times 10^{16}$	Roentgen	When highly energetic electrons are stopped by a heavy metal target of high melting point	By the fluorescence produced on a zinc sulphide screen. The photographic film gets affected.
Ultraviolet	$3 \times 10^{16} - 7.5 \times 10^{14}$	Ritter	Sunlight, arc-lamp or spark	By their chemical activity on dyes. Photographic plates get affected. It causes fluorescence.

Visible light	$7.5 \times 10^{14} - 3.75 \times 10^{14}$	Newton	Sunlight, light from electric bulb, flame, white hot bodies	Other objects can be seen in its presence.
Infrared waves	$3.75 \times 10^{14} - 3 \times 10^{12}$	Herschel	Lamp with thoriated filament, heated silicon carbide rod, red hot bodies	Heating effect is more. The mercury rises rapidly when a thermometer with the blackened bulb is kept in these radiations.
Microwaves	$3 \times 10^{12} - 3 \times 10^9$	Hertz	Electronic devices such as crystal oscillators	Oscillatory electrical circuit
Radio waves	Below 3×10^9	Maxwell predicted the existence of radio waves.	TV and radio transmitters	Aerials of radio and TV receiver

□ **Properties common to all electromagnetic spectrum:**

(i) The electromagnetic waves of the entire wavelength range do not require any material medium for their propagation.

(ii) They all travel with the same speed in vacuum, which is the same as the speed of light in vacuum, i.e., 3×10^8 m/s.

(iii) They exhibit the properties of reflection and refraction.

(iv) These waves are not deflected by the electric and magnetic fields.

(v) These waves are transverse waves.

□ **Properties and uses of the electromagnetic spectrum:**

➤ **γ-rays**

These rays are the most energetic electromagnetic waves of wavelength less than 0.01 nm.

□ **Properties of γ-rays are as follows:**

(i) They cause fluorescence, when they strike on the fluorescent materials as zinc sulphide.

(ii) They can easily penetrate through the thick metallic sheets.

(iii) γ-rays can easily penetrate through the human body.

□ **Uses of γ-rays are as follows:**

(i) They are used in the treatment of cancer and tumours.

(ii) γ-rays are used to preserve the foodstuffs for a long time.

(iii) It provides valuable information about the structure of the atomic nucleus.

➤ **X-rays**

X-rays, discovered by German physicist W Roentgen, having a range of wavelength from 0.01 to 10 nm.

They are produced when highly energetic cathode rays are stopped by a heavy metal target of high melting point.

□ **Properties of X-rays are as follows:**

(i) They have high penetrating power.

(ii) They strongly affect the photographic plate.

(iii) They cause fluorescence in certain material such as zinc sulphide, etc.

□ **Uses of X-ray are as follows:**

(i) For detection of fracture, foreign bodies like bullets, stone in a human body, etc.

(ii) For detecting faults, cracks, flaws and holes in the final product of metals

(iii) For curing untraceable skin diseases and malignant growths

(iv) For investigation of the structure of crystals, arrangement of atoms, etc.

➤ **Ultraviolet Radiation**

It was discovered by Ritter in 1801. They are produced by some special lamps and very hot bodies. Ultraviolet rays coming from the Sun are absorbed by the ozone layer in the Earth's atmosphere. The wavelength range varies from 10 to 400 nm.

□ **Properties of ultraviolet radiation are as follows:**

(i) They can pass through quartz but are absorbed by glass.

(ii) They can be scattered by dust particles in the atmosphere.

(iii) They cause health hazards like skin cancer, if our body is exposed for a long period of time.

□ **Uses of ultraviolet radiation are as follows:**

(i) For sterilising purposes

(ii) For detecting the purity of gold, eggs, ghee, etc.

(iii) For producing vitamin D in food of plants and animals

➤ **Visible Light**

It is the narrow region of the electromagnetic spectrum, which can be detected by the human eyes. Its wavelength ranges from 390 to 700 nm.

➤ **Uses of visible light are as follows:**

(i) The visible light emitted or reflected from the object around us provides the information surrounding us.

(ii) It is used in photography, photosynthesis and to see objects around us.

➤ **Infrared Radiation**

It was discovered by Herschel. They are sometimes called heat wave, because their absorption causes the heating effect in the bodies and surroundings. They are produced by hot bodies and molecules. Its range is from 700 nm to 1 mm.

□ **Properties of infrared radiation are as follows:**

(i) They do not affect ordinary photographic film.

(ii) They are absorbed by the glass but are not absorbed by rock salt.

Light

(iii) They are detected by their heating property.

(iv) They are less scattered by the atmosphere.

□ **Uses of infrared radiation are as follows:**

(i) They are used in photography at night and also in mist and fog.

(ii) They are used for therapeutic purpose.

(iii) They are used in remote control of TV and other gadgets.

(iv) They are used as signals during the war.

➤ **Microwaves:**

Microwaves are the electromagnetic waves of the wavelength having a range of 10^{-3} m to 0.3 m.

They are produced by special vacuum tubes.

□ **Uses of microwaves are as follows:**

(i) They are used for aircraft navigation.

(ii) They are used for cooking purposes.

(iii) They are used for observing the movement of trains, while sitting in a microwave operated control room.

➤ **Radiowaves:**

These are the waves of longest wavelength amongst all the electromagnetic waves. They have a wavelength above 0.1 m. They show all the properties of electromagnetic waves.

□ **Uses of radiowaves are as follows:**

(i) It is used in ground wave propagation.

(ii) It is used in sky wave propagation.

(iii) They are used mainly in radar communication and television transmission.

➤ **Scattering of light:** The process of scattering is absorption of light by the tiny particles followed by re-radiation in different directions.

➤ **Tyndall effect:** The phenomenon of scattering of light by the colloidal particles.

➤ The molecules of air and other fine particles in the atmosphere are more effective in scattering light of shorter wavelengths at the blue end than light of longer wavelengths at the red end. Thus, when sunlight passes through the atmosphere, the fine particles in air scatter the blue colour (shorter wavelengths) more strongly than red. The scattered blue light enters our eyes. So, we see the colour of sky as blue.

➤ In absence of atmosphere (in moon, in deep space etc.), no scattering takes place and the sky appears dark.

➤ The 'danger' signal lights are red in colour since the red is least scattered by rain, fog or smoke.

MNEMONICS

Concept: Thicker lens has shorter focal length.

Mnemonics: The Loaf is Soft and Fresh.

Interpretation:

T: Thicker

L: Lens

S: Softer

F: Focal length



Sound

TOPIC-1: REFLECTION OF SOUND – ECHO, RESONANCE

Concepts Covered: Reflection of sound waves, Echoes, Natural vibrations, Damped vibrations, Forced vibrations, Resonance



Revision Notes

➤ Reflection of sound waves:

[Board 2022 sem 1]

The returning back of the sound wave on striking a surface such as wall, metal sheet, etc., is known as reflection of sound wave. It does not require a smooth and shining surface like mirror. The reflection of sound takes place in accordance with the same laws those governing the reflection of light. The condition for reflection of sound wave is that the size of the reflecting surface must be bigger than the wavelength of the sound wave.

➤ Echo:

It is a reflection of sound, arriving at the listener after the sensation of original sound died out.

Sound persists in human brain for $\frac{1}{10}$ s.

So, reflected sound should reach after that time. Example: The echoes are produced by the bottom of a well, by a building or by the walls of an enclosed room and an empty room.

[Board 2024, 2022 sem 1]

➤ Applications of Echoes

➤ Bats and dolphins make use of the phenomenon of echoes of ultrasonic waves in nature.

➤ Trawlerman makes use of echoes for finding the depth of ocean beds or for detecting submerged objects.

➤ RADAR and SONAR also make use of echoes for finding the position and distance of an enemy airplane, under water dangers.

[Board 2024]

➤ A tuning fork is made by shaping a metal piece in the form shown alongside. It enables us to produce a pure sound note.

➤ Conditions for formation of echo/hearing the echo distinctly:

(i) The size of the obstacle/reflector must be large compared to the wavelength of the incident sound (for reflection of sound to take place).

(ii) The distance between the source of sound and the reflector should be atleast 17 m (so that the echo is heard distinctly after the original sound is over).

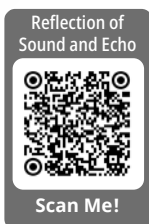
(iii) The intensity or loudness of the sound should be sufficient for the reflected sound reaching the ear to be audible. The original sound should be of short duration.

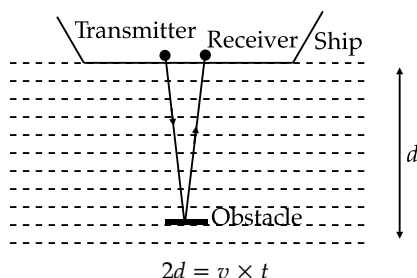
➤ Echoes also find use in medical field for imaging of human organs (womb, liver, uterus).

➤ Echoes find application in **SONAR (Sound navigation and ranging)**. In order to find the distance of obstacle from ship, ultrasonic waves are transmitted and then reflected waves are received by the receiver.

Let the distance of the obstacle from source of sound be “d” then,

[Board 2024]





Where,

v is the velocity of ultrasonic waves in water and t is the time between sending and receiving of waves.

➤ The vibration of a body when it is disturbed from its equilibrium position and allowed to vibrate freely, without any external driving or damping force acting on it is known as **natural vibration**.

➤ The periodic vibrations of a body of decreasing amplitude in presence of a resistive force are called the **damped vibrations**. Examples: oscillations of simple pendulum in air.

➤ The periodic vibrations of a body of constant amplitude in the absence of any external force on it, are called the **free vibrations**. Examples: vibration in instruments like sitar and violin when they are plucked.

➤ The vibrations of a body which take place under the influence of an external periodic force acting on it are called the **forced vibrations**.

➤ The frequency of vibrations of a body executing forced vibrations equals to the frequency of the applied periodic force. Examples: vibrations produced in the diaphragm of microphone sound box with frequency matching the speech of speaker. [Board 2024]

➤ **Resonance** is a special case of forced vibrations. When the frequency of an externally applied periodic force on a body is equal to its natural frequency, the body readily

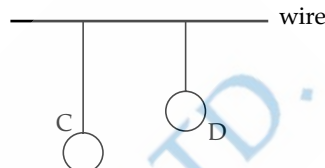
begins to vibrate with an increased amplitude. This phenomenon is known as resonance. The vibrations of large amplitude are called the resonant vibrations. For example, tuning of a radio is based on resonance.

➤ Condition for resonance:

Resonance occurs only when the frequency of the applied force is exactly equal to the natural frequency of the vibrating body.

Example 1

Two pendulums C and D are suspended from a wire as shown in the figure given below. Pendulum C is made to oscillate by displacing it from its mean position. It is seen that D also starts oscillating.



- (i) Name the type of oscillation, C will execute.
- (ii) Name the type of oscillation, D will execute.
- (iii) If the length of D is made equal to C, then what difference will you notice in the oscillations of D?
- (iv) What is the name of the phenomenon when the length of D is made equal to C? [ICSE 2019]

Ans. (i) Free vibration in vacuum / damped vibrations when a medium is present

(ii) Forced vibrations

(iii) Amplitude of vibration of pendulum D increases substantially.

(iv) Resonance.

[ICSE Marking Scheme, 2019]

➤ Difference between the free and damped vibrations:

S. No.	Free vibrations	Damped vibrations
1.	The amplitude of free vibrations remains constant and the vibrations continue forever.	The amplitude of damped vibrations gradually decreases with time and ultimately the vibrations cease.
2.	There is no loss of energy in free vibrations.	In each vibration, there is some loss of energy in the form of heat.
3.	No external force acts on the vibrating body. The vibrations are only under the restoring force.	The frictional or damping force acts to oppose the motion.
4.	The frequency of vibrations is nearly equal to the natural frequency and it remains constant.	The frequency of vibrations is less than the natural frequency.

➤ Difference between the free and forced vibrations:

S. No.	Free vibrations	Forced vibrations
1.	The vibrations of a body in absence of any resistive force are called the free vibrations.	The vibrations of a body in presence of an external periodic force are called the forced vibrations.
2.	The frequency of vibration depends on the shape and size of the body.	The frequency of vibration is equal to the frequency of the applied force.
3.	The frequency of vibration remains constant.	The frequency of vibration changes with change in the frequency of the applied force.
4.	The amplitude of vibration is constant.	The amplitude of vibration depends on the frequency of applied force.

➤ Difference between the forced and resonant vibrations:

S. No.	Forced vibrations	Resonant vibrations
1.	The vibrations of a body under an external periodic force of frequency different from the natural frequency of the body are called the forced vibrations.	The vibrations of a body under an external periodic force of frequency exactly equal to the natural frequency of the body are called the resonant vibrations.
2.	The amplitude of vibration is usually small.	The amplitude of vibration is very large.
3.	The vibrations of the body are not in phase with the external periodic force.	The vibrations of the body are in phase with the external periodic force.
4.	These vibrations last for a very small time after the periodic force has ceased to act.	These vibrations last for a long time after the periodic force has ceased to act.

MNEMONICS**Mnemonics:** Dr. HOPI**Interpretation:****D:** Diagnosis (ultrasound)**O:** Observation (bats, dolphins)**R:** Remote sensing (SONAR)**P:** Predict earthquakes (seismic waves)**H:** Hearing**I:** Inspection (cracks in pipes)

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TOPIC-2: CHARACTERISTICS OF SOUND

Concepts Covered: Loudness, pitch, quality, sound level in dB and noise pollution



Revision Notes

➤ Characteristics of Sound Waves

Sound waves can be described by its:

● Wavelength

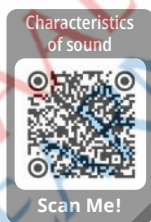
The distance between the two consecutive compressions (C) or two consecutive rarefactions (R), is called the wavelength. In other words, the distance between two consecutive crests or two consecutive troughs of sound wave is called wavelength. Wavelength is the minimum distance in which sound wave repeats itself. Its SI unit is metre (m). It is denoted by ' λ '.

● Frequency

The number of complete waves (or oscillations) produced in one second is called frequency of the wave. It is the number of vibrations that occurs per second. The frequency of a wave is fixed and does not change even when it passes through different substances. Its SI unit is hertz (Hz). It is denoted by ' v ' or ' f '.

● Time Period

The time taken by two consecutive compressions or rarefactions to cross a fixed point is called the time period of the sound wave. In other words, the time required to produce one complete wave (or oscillations) is called time period of the wave. It is denoted by symbol T . Its SI unit is second (s). The time period of a wave is the reciprocal of its frequency. *i.e.*,



$$\text{Time period} = \frac{1}{\text{Frequency}} \quad \text{Or} \quad T = \frac{1}{v}$$

$$\text{Or} \quad \text{Frequency} = \frac{1}{\text{Time period}} = \frac{1}{T}$$

● Amplitude

The maximum displacement of the particles of the medium on either side from their original mean positions on passing a wave through the medium, is called amplitude of the wave. It is used to describe the size of the wave. It is usually denoted by A . Its SI unit is metre (m). The amplitude of a wave is the same as the amplitude of the vibrating body producing the wave.

● Speed

The distance travelled by a wave in one second is called speed of the wave or velocity of the wave. It is represented by letter ' v '.

Its SI unit is metre per second (ms^{-1}). [Board 2022 sem 2]

➤ Relationship among speed, frequency and wavelength of a wave

Suppose distance travelled by a wave is λ (wavelength) in time T ; then the speed is given by

$$v = \frac{\lambda}{T}$$

$$\text{We know that, } f = \frac{1}{T}$$

[Board 2020]

$$\text{Therefore, } v = \lambda \times f \text{ or } (v = f\lambda)$$

Or, Speed (or velocity) = Frequency $\times$

Wavelength

- Three main characteristics of a given musical sound are (i) loudness, (ii) pitch and (iii) quality or timbre.
- **Loudness** is the property by virtue of which a loud sound can be distinguished from a faint one, both having the same pitch and quality. **[Board 2024]**
- Loudness is proportional to the square of amplitude of the wave. **[Board 2024]**
- **Pitch** is that characteristic of sound by which an acute (or shrill) note can be distinguished from a grave or flat note. **[Board 2022 sem 2]**

➤ Pitch of a note depends on its frequency. Two notes sounded on the same instrument with same amplitude, will differ in pitch when their vibrations are of different frequencies.

➤ **Quality (or timbre)** of a sound is that characteristic which distinguishes the two sounds of the same loudness and same pitch, but emitted by two different instruments.

➤ The quality of a musical sound depends on the waveform.

Thus the quality of a musical sound depends on the number of the subsidiary notes and their relative amplitudes present along with the principal note.

S. No.	Musical sound	Noise
1.	It is pleasant, smooth and acceptable to the ear.	It is harsh, discordant and non-acceptable to the ear.
2.	It is produced by the vibrations which are periodic.	It is produced by an irregular succession of disturbances.
3.	All the component waves are similar without any sudden change in their wavelength and amplitude.	The component waves change their character suddenly and they are of short duration.
4.	The sound level is low (between 10 dB and 30 dB).	The sound level is high (above 120 dB).
5.	The waveform is regular. Example: The sound produced by the musical instruments.	The waveform is irregular. Example: The sound produced by an aeroplane, road roller, industrial machines, etc.

MNEMONICS

Concept: Relation of frequency and wavelength of sound with its velocity. $v = \text{frequency } \lambda$.

Mnemonics: Seagull Egg is Famous in Malaysia and Washington.

Interpretation:

S: Speed

E: Equal to

F: Frequency

M: Multiplied by

W: Wavelength



Current Electricity

TOPIC-1: Ohm's law

Concepts Covered: Ohm's law, Potential difference and EMF, Resistance and their combination in series and parallel, Internal resistance



Revision Notes

➤ A sustained electric current flows through a conductor only when it is connected to a source of EMF.

➤ The electric current in a given conductor is simply the rate of flow of charge across its cross-section. Thus,

$$I = Q/t$$

➤ The unit of current is the **ampere** (A).

$$1 \text{ A} = 1 \text{ coulomb per second} = 1 \text{ Cs}^{-1}$$

➤ The current **flowing** through a wire, is said to be one ampere when **one** Coulomb of charge flows across its cross-section in **one** second.

➤ Potential is the electrical state of a conductor that determines the direction of flow of charge when two conductors are either kept in contact or joined by a metallic wire. The potential at a point is defined as the amount of work done in bringing a unit positive charge from infinity to that point. The potential difference (p.d.) between two points is simply the work done in transporting a positive charge of one coulomb from the first point to the second point. $V = W/Q$, i.e., $W = Q.V$

➤ The unit of potential difference is **Volt** (V). We have $1 \text{ volt} = 1 \text{ joule per coulomb} = 1 \text{ JC}^{-1}$. The potential difference between two points equals one volt if the work done in transporting a charge of one coulomb between these points equals one joule.

➤ **Electromotive Force:** When no current is drawn from a cell, then the potential difference between its terminals is called the electromotive force (e.m.f.). It is denoted by E . Its unit is Volt (V). The electromotive force of a cell depends upon

(i) the material of the electrodes.

(ii) the electrolyte used in the cells.

➤ **Ohm's Law :**

The current flowing through a given conductor is directly proportional to the potential difference across its ends, provided the physical conditions of the conductor (e.g., its length, its area of cross-section, its temperature, etc.) remain constant.

Ohm's law



Scan Me!

[Board 2024]

$$I \propto V$$

Or, $I = GV$ (where G is proportionality constant and is known as the conductance of the conductor. $G = \frac{1}{R}$, where R is the resistance of the conductor)

$$\text{Or, } I = \frac{V}{R}$$

$$\therefore V = IR$$

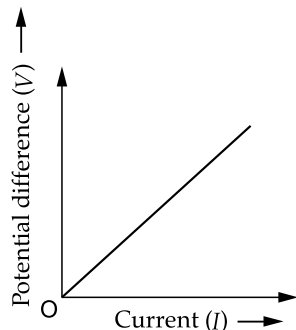
The obstruction offered to the flow of current in a conductor (or wire) is called its resistance. The unit of resistance is **ohm** (Ω).

The resistance of a conductor equals the ratio of the potential difference across its ends to the current flowing through it.

A conductor has a resistance of one ohm when a potential difference of one volt across its ends causes a current of one ampere to flow through it.

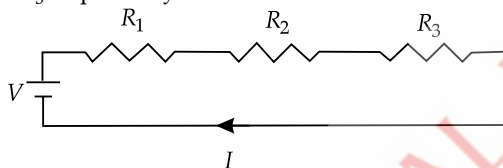
➤ **V-I graph :**

The below graph gives the graphical representation of Ohm's law where the slope of V-I graph gives the resistance of the conductor.



➤ **Combination of resistors in series :**

Three resistors R_1 , R_2 and R_3 are connected in series. The same current I , flows through all three resistors but the potential across each resistor is different and let it be V_1 , V_2 and V_3 respectively.



$$V = V_1 + V_2 + V_3$$

$$= IR_1 + IR_2 + IR_3$$

where V is the applied potential difference

$$IR_{eq} = I(R_1 + R_2 + R_3)$$

$$R_{eq} = R_1 + R_2 + R_3$$

So, the equivalent resistance in a series combination is the sum of all resistances. [Board 2024]

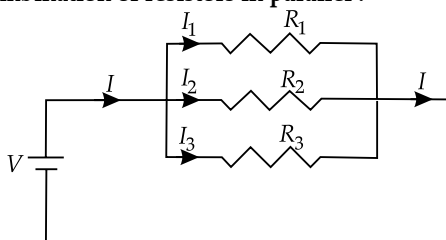
➤ The following two facts are to be noted about the series combination.

(i) The current has a single path for its flow. Hence, the same current passes through each resistor, so the potential difference across any resistor is directly proportional to its resistance.

(ii) The potential difference across the entire circuit is equal to the sum of potential differences across the individual resistor, i.e.,

$$V = V_1 + V_2 + V_3 \dots$$

➤ **Combination of resistors in parallel :**



Three resistors R_1 , R_2 and R_3 are connected in parallel. The same potential difference, V exist through all the three resistors but the current across each resistor is different and let it be I_1 , I_2 and I_3 respectively.

$$\text{Net current, } I = I_1 + I_2 + I_3$$

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

So, the reciprocal of equivalent resistance in a parallel combination is equal to the sum of the reciprocal of individual resistances.

➤ The following two points are to be noted about the parallel combination.

(i) The potential difference across each resistor is the same (say $V_1 = V_2 = V_3$), which is equal to the potential difference across the terminals of the battery (or source).

(ii) The main current I from the battery divides itself into different arms. The low resistance arm allows more current and the high resistance arm allows less current, i.e., the current in a resistor is inversely proportional to its resistance. The sum of currents I_1 , I_2 , I_3 ... in the separate branches of the parallel circuit is equal to the current I drawn from the source, i.e., $I = I_1 + I_2 + I_3 + \dots$

➤ When the resistance of the combination is to be increased, they are connected in series and when the resistance of the combination is to be decreased, in order to pass a heavy current in the circuit, they are connected in parallel.

➤ **Factors affecting the resistance of a conductor**

The electrical resistance of a conductor depends on the following factors :

(i) **Length of the conductor :** The resistance of a conductor R is directly proportional to its length l ,

$$\text{i.e., } R \propto l \quad \dots\dots(i)$$

∴ When the length of a wire is doubled or halved, then its resistance also gets doubled or halved respectively.

(ii) **Area of cross-section of the conductor :** The resistance of a conductor R is inversely proportional to its area of cross-section A .

$$\text{i.e., } R \propto \frac{1}{A} \quad \dots\dots(ii)$$

∴ When the area of cross-section of wire is doubled, its resistance gets halved and if area of cross-section of wire is halved, its resistance will get doubled.

From the above relation (i) and (ii), we can write $R \propto \frac{l}{A}$ or $R = \rho \frac{l}{A}$ where, ρ is the constant of proportionality called resistivity or specific resistance of the conductor. It is a material dependant constant.

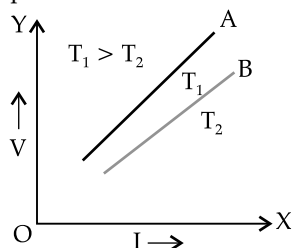
➤ Resistivity of a material is the resistance of a wire of that material of a unit length and unit area of cross section.

Current Electricity

(iii) **Nature of the material of the conductor :** The resistance of a conductor depends on the nature of the material it is made of. Some materials have low resistance, while others have high resistance.

(iv) **Effects of temperature :**

(a) Resistance of a conductor increases linearly with increase in temperature.

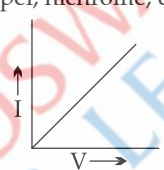
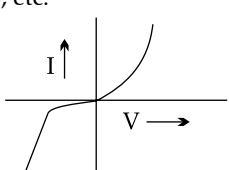


(b) Resistance of a semiconductor decreases with increase in temperature.

(c) Resistance of insulators (non-conductor) decreases with increase in temperature.

(d) Resistance of electrolytes decreases with increase in temperature.

➤ **Difference between Ohmic and Non-ohmic resistor :**

S. No.	Ohmic resistor	Non-ohmic resistor
1.	It obeys Ohm's law, i.e., V/I is constant for all values of V or I .	It does not obey the Ohm's law, i.e., V/I is not the same for all values V or I .
2.	The graph for potential difference (V) versus current (I) is a straight line.	The graph of potential difference (V) versus current (I) is not a straight line.
3.	The slope of $V-I$ graph is same for all values of V or I at a given temperature. <i>Examples :</i> All metallic conductors such as silver, iron, copper, nichrome, etc.	The slope of $V-I$ graph is different at different values of V or I at a given temperature. <i>Examples:</i> Junction diode, diode valve, transistor, filament of a bulb, etc.
	 Ohmic (Aluminium)	 non-ohmic (Semiconductor diode)

➤ **Difference between Resistance and Resistivity :**

S. No.	Resistance	Resistivity
1.	Resistance of a conductor is the obstruction offered by the conductor in the flow of current through it. It is measured by the potential difference needed across the conductor to flow 1 ampere current through it. $R = \frac{V}{I}$	It is the property of the conductor due to which it offers resistance to the flow of current through it. It is measured by the resistance offered by 1 m length of wire of that material of area of cross-section 1 m^2 . $\rho = \frac{RA}{L}$
2.	The resistance of a conductor depends upon its material, temperature, length and area of cross section.	The resistivity of a conductor depends only on its material and temperature but not on its shape and size.
3.	Its SI unit is ohm (or Ω) .	Its SI unit is ohm metre (or $\Omega \text{ m}$) .

Resistivities of substances at room temperature

Substances	$\rho(\Omega \text{ m})$
CONDUCTORS	
Silver	1.6×10^{-8}
Copper	1.62×10^{-8}
Aluminium	2.63×10^{-8}
ALLOYS	
Nichrome	100×10^{-6}
INSULATORS	
Glass	$10^{10} - 10^{14}$
Wood	$10^5 - 10^{18}$

➤ **Superconductor :** It is a substance of zero resistance at a very low temperature. e.g., Mercury, niobium, etc.

Ohmic and non-ohmic resistor :

The conductor which obeys Ohm's law is called ohmic resistor (or linear resistance), e.g., Silver, nichrome, copper, iron, etc. The graph of V vs I is a straight line. The conductor which does not obey Ohm's law is known as non-ohmic resistor (for non-linear resistance), e.g., Triode valve, junction diode, transistor, etc. The graph of V vs I is a curved line.

- The p.d. of a cell is defined as the energy spent (or the work done) per unit charge in taking a positive charge around the complete circuit of the cell (i.e., in the circuit outside the cell as well as in the electrolyte inside the cell).
- When no current is drawn from a cell, i.e., when the cell is in open circuit, the potential difference between the terminals of the cell is called its **electromotive force** or (EMF).

➤ **Difference between EMF and Terminal voltage of cells :**

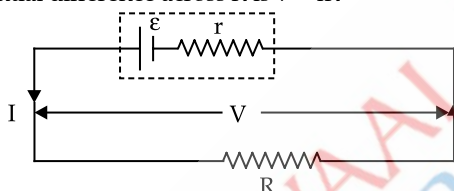
S. No.	EMF of cell	Terminal voltage of cell
1.	It is measured by the amount of work done in moving a unit positive charge in the complete circuit inside and outside the cell.	It is measured by the amount of work done in moving a unit positive charge around the complete circuit inside and outside the cell.
2.	It is the characteristic of the cell, i.e., it does not depend on the amount of current drawn from the cell.	It depends on the amount of current drawn from the cell. More the current is drawn from the cell, less is the terminal voltage.
3.	It is equal to the terminal voltage when cell is not in use, while greater than the terminal voltage when a cell is in use.	It is equal to the EMF of the cell, when the cell is not in use, while less than the EMF when the cell is in use.

- The resistance associated with the electrolyte of a cell—within its electrodes—is known as the **internal resistance** of the cell.

- When a cell of emf ϵ , and internal resistance r , is connected across an external resistance R , current flows through the circuit, which is given by

$$I = \epsilon / (R + r)$$

and Potential difference across R is $V = IR$



- The internal resistance of a cell depends on
 - (i) the nature, concentration and temperature of its electrolyte.
 - (ii) the surface area of the ('dipped within the electrolyte' part) electrodes.
 - (iii) the distance between the electrodes.

- An increase in the surface area of the electrodes causes the internal resistance to decrease, whereas an increase in the distance between the electrodes causes the internal resistance to increase. Internal resistance decreases with increasing temperature or concentration of an electrolyte.

Example 1

How many protons will constitute a charge of 1 C?

[OEB]

Ans. We know that

$$Q = ne$$

where, $Q = 1 \text{ C}$

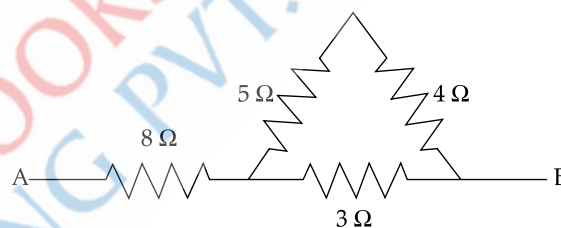
$$e = 1.6 \times 10^{-19} \text{ C (charge on proton)}$$

$$\Rightarrow n = \frac{Q}{e} = \frac{1}{1.6 \times 10^{-19}}$$

$$\Rightarrow n = 6.25 \times 10^{18} \text{ Protons}$$

Example 2

Calculate the effective resistance across AB.



[ICSE 2019]

Ans. 5Ω and 4Ω resistors are in series. So, the equivalent resistance $R_1 = 9 \Omega$.

R_1 and 3Ω are in parallel. So, equivalent resistance

$$= R_2 = \frac{3 \times 9}{3 + 9} = 2.25 \Omega$$

R_2 and 8Ω are in series, So, the effective resistance = $R_{\text{eff}} = 10.25 \Omega$.

MNEMONICS

Concept: Equivalent resistance of series and parallel combination of resistors.

Mnemonics: Sunny Arrived Dadar and Pari Arrived Rajasthan.

Interpretations:

S: Series combination

A: Add

D: Directly

P: Parallel combination

A: Add

R: Reciprocals

(For series combination of resistors:

$$R_s = R_1 + R_2 + R_3 + \dots$$

For parallel combination of resistors:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots)$$

TOPIC-2: ELECTRIC POWER AND ENERGY, FUSE, HOUSEHOLD CIRCUITS

Concepts Covered: Electrical power, Electrical energy, Household circuits, Safety precaution



Revision Notes

➤ **Electrical Energy** is a form of energy resulting from the flow of electric charge. It is one of the most versatile and widely used energy forms in modern life. **Electrical Energy** is utilised and converted into other forms of energy. It powers appliances, machines, lighting, communication systems and more. For example, heater converts electrical energy into heat energy, motor converts electrical energy into mechanical energy, electric lamp converts electrical energy into light energy (and some heat energy), loudspeaker converts electrical energy into sound energy and many more.

➤ **Electric Power** is the rate at which electrical energy is transferred or consumed or electric work is done in a circuit. It indicates how much energy is used by an electrical device per unit of time.

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

$$P = \frac{W}{t}$$

By the definition of potential difference, work needed to move a charge Q through a potential difference V is

$$W = QV$$

$$\text{But } Q = I \times t$$

$$W = VIt$$

Using Ohm's law, $V = IR$

Or

Electric power = Voltage $\times$ Current

SI unit of power is "watt".

1 watt is the electric power consumed when a current of 1 ampere flows through a circuit having a potential difference of 1 volt.

(Board 2024)

➤ **SI Units :**

S. No.	Physical Quantity	SI Unit	
		Name	Symbol
1.	Current	ampere	A
2.	Voltage	volt	V
3.	Resistance	ohm	Ω
4.	Heat energy/Electric energy	joule	J
5.	Electric power	watt	W

➤ The unit often used for estimating the consumption of electrical energy is the kilowatt hour (kWh). One **kilowatt-hour** (kWh) is the electrical energy consumed by an electrical appliance of power 1 kW when it is used for 1 h.

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} \\ = 3.6 \times 10^6 \text{ J}$$

➤ **Electrical energy consumed by appliance** (in kWh) = power (in kW) $\times$ time (in h)

$$= \text{power (in watt)} \times \text{time (in hour)} \div 10^3$$

$$= V \text{ (volt)} \times I \text{ (in ampere)} \times t \text{ (hour)} \div 10^3$$

➤ **Cost of electricity** = Electrical energy in kWh $\times$ cost per kWh

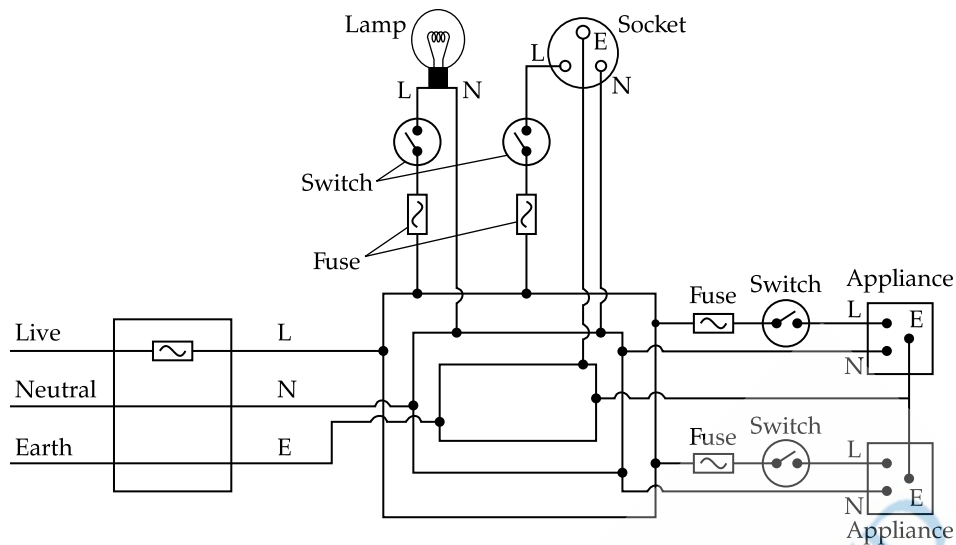
Some house hold electrical appliances with their power rating at 240 V.

Appliances	Power (W)
Filament Bulb	60-200
CFL	15-40
Electric Heater	1500-2000
Electric Iron	750-1000
TV	50-300
Fan	33-100
AC	500-4000

➤ In India, electric power for domestic use is supplied with alternating current (a.c.) at a frequency of 50 Hz with a (rms) potential difference of 220 V between the phase (or live) wire and the neutral wire.

➤ The neutral and the earth wires of the local sub-station are usually connected together, so that the neutral wire is at the same potential as the earth. The live or the phase wire is then at a potential of 220 V (rms) with respect to either the neutral or the earth wire.

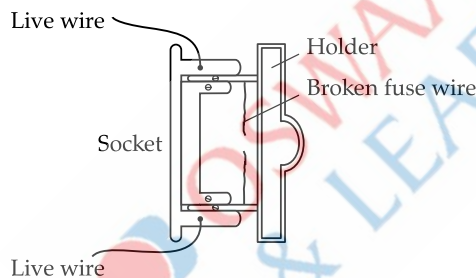
➤ **Household Circuits:** We use two systems for domestic wiring – the tree system and ring system. Of the two, the ring system is more popular. The ring system is a common wiring system for homes that involves running wires from the main fuse box around the house and back to the fuse box. The system is named for its ring-like shape. It is shown as



➤ **Advantages of the ring system :** The ring system has the following advantages :

(i) **In the ring system**, the current from the mains can reach an individual appliance through two separate paths. Thus, each appliance gets connected to the mains effectively through a thick wire. Therefore, the wire required for the main ring is of a lower current-carrying capacity than that which would be required for a direct connection to the mains. This reduces the cost of wiring considerably.

(ii) **Each appliance has a separate fuse.** Therefore, if due to some fault, the fuse of one appliance burns, it does not affect the other appliances.



(iii) In this system, all the plugs and sockets of the same size can be used, but each socket should have its fuse of a rating suitable for the appliance to be connected with it.

(iv) While installing a new appliance in a room, a new line up to the distribution box is not required, but it can be directly connected to the ring circuit in that room. The care is taken that the total current drawn from the mains in the ring circuit does not exceed the rating of the main fuse (viz. 30 A).

➤ A fuse is a safety device that limits the flow of current in a circuit and thus protects the device from being damaged.

➤ Fuse is made up of material having high resistivity and low melting point.

➤ A switch is an ON/OFF device that is connected in a circuit to the live wire.

➤ In modern electrical systems, MCBs (**Miniature Circuit Breakers**) are more commonly used than fuses due to their advantages in safety, convenience, and reliability. An MCB is a safety device used in electrical circuits to prevent damage caused by overloads or short circuits. It automatically switches off the electrical supply when it detects excessive current flow, protecting appliances and the wiring from potential hazards like fire or damage. Comparison between Fuse and MCB.

	Fuse	MCB
Function	Protects by melting when current exceeds the rated limit.	Protects by tripping (disconnecting) the circuit.
Reusability	Not reusable; requires replacement after blowing	Reusable; can be reset after tripping.

➤ Earthing protects the appliance and the user from getting damaged/ hazardous shock due to short circuiting. If excess current due to short circuit flows through the earth wire to the earth. In local earthing, the earth connection at a meter is connected to a thick copper wire, which is buried deep in the earth and its ends are connected to a copper plate surrounded by a mixture of charcoal and salt. The thick copper wire is covered by a hollow pipe and water is poured into the pipe from time to time to provide a conducting layer between the plate and the ground.

➤ **Colour coding of wires in a cable :**

Wire	Colour	
	old convention	new convention
Live	Red	Brown
Neutral	Black	Light blue
Earth	Green	Green or Yellow

Current Electricity

➤ We need both **conductors** (materials that let electricity flow through them easily) and **insulators** (materials that do not let electric current flow through them easily) for safe and proper use of electrical energy.

➤ All electric circuits (closed conducting paths used for ensuring flow of electric current) need :

(i) a source of electricity (cell, battery, generator, 'mains', etc.)

(ii) conducting wires or conductors for providing a path for the flow of current and

(iii) a switch or a key for switching 'ON' and 'OFF' electric energy as per our requirements.

➤ **Connection of all appliances (bulb, fan, socket, etc.) with the mains :**

We note that all the electrical appliances (say, bulbs, fans, sockets, etc.) are connected in parallel with the mains, each with a separate switch and a separate fuse connected with the live wire.

➤ **Advantages of connecting the appliances in parallel :**

(i) Each appliance gets connected to 220 V supply for its normal working.

(ii) Each appliance works independently without being affected by the status of the other appliance being switched on or off.

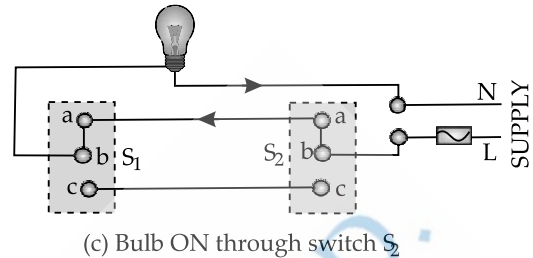
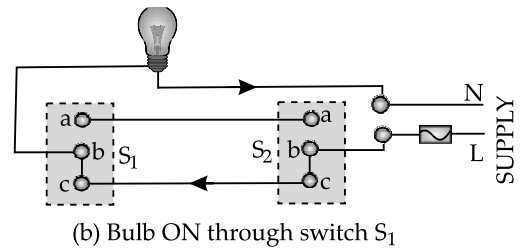
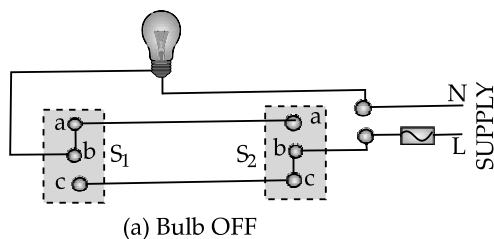
➤ **Disadvantages of connecting the appliances in series:**

(i) The voltage of source gets divided in all the appliances connected in series, in a ratio of their resistances, so each appliance does not operate at its rated voltage.

(ii) On connecting one more appliance in the same circuit, the resistance of the circuit will increase. Hence, it will reduce the current in the circuit, so each appliance will get less power.

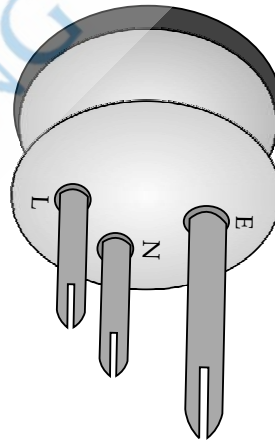
(iii) All appliances connected in series operate simultaneously. None of the appliances can be operated independently. If one appliance is switched off or not operated, no other appliance connected with it in series will then operate.

➤ **Staircase wiring** or dual control switch has two pathways for making the switch ON/OFF. The following diagrams illustrate a dual switch with two switches S_1 and S_2 . The circuit can be switched on either using switch S_1 or S_2 . When the connection ba is changed to bc by switch S_1 , the circuit is completed and the bulb glows. When the connection bc is changed to ab by switch S_2 , the circuit is incomplete and the bulb goes off.

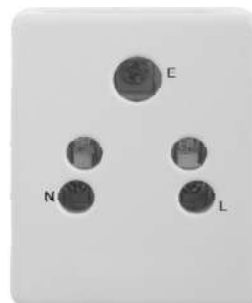


Similarly the bulb can be switched ON by S_2 and switched OFF by S_1 .

➤ Three pin plug has three pins – the top pin is for earth (E), the left pin is for live (L) and the right pin is for neutral (N). The earth pin is made longer and thicker so that firstly earth connection is made and if the device is faulty then after live connection is made, the current passes to the earth through earth wire.



➤ A socket is fixed in the board of the circuit so that plug can be inserted. It has three holes which have a hollow metallic tube. The upper bigger hole is for earth connection, the left is for neutral and right for a live wire.



➤ **Safety precautions while using electricity :**

- (i) Never touch the appliance with wet hands.
- (ii) The wires carrying current should be having higher capacity than the total current which would flow when all the appliances are switched on.
- (iii) Each appliance must be provided with a fuse and earthing.
- (iv) The insulation of wires should be of good quality and that should be checked from time to time.
- (v) Local earthing should be done near kWh metre.

Properties of the material to be used in electrical circuits for different purpose.

MATERIALS	PROPERTIES
Fuse Wire	High Resistivity, Low Melting Point
Connecting Wire	Low Resistivity
Filament Bulb	Low Resistivity, High Melting Point
Heating Coil	High Resistivity, High Melting Point

Example 3

In a house hold circuit there is 5 A main fuse in 220 V supply line. Maximum how many 100 W lamps can be lighted simultaneously?

- (a) 5
- (b) 10
- (c) 11
- (d) There is no maximum limit

Ans. Option (c) is correct.

Explanation: Each 100 W lamp draws $\frac{100 \text{ W}}{220 \text{ V}}$
= 0.45 A current.

So, maximum number of lamps = $\frac{5 \text{ A}}{0.45 \text{ A}}$

= 11.11

So, 11 100 W lamps can be lighted simultaneously.

MNEMONICS

Concept: Formula of electrical energy.

Mnemonics: Esha Prefers Everyday to Come to School at Right Time.

Interpretations:

E: Electrical energy

E: Equals to

P: Product of

C: Current

S: Square

R: Resistance

T: Time

$W = I^2 R t$



Magnetism

TOPIC-1: MAGNETIC EFFECT OF CURRENT

Concepts Covered: Oersted's experiment field lines due to current in a straight wire, Right hand thumb rule – magnetic field due to a current in a loop, Electromagnets and permanent magnets, Fleming's left hand rule, DC motor



Revision Notes

➤ Besides the *heating effect* and the *chemical effect*, current also has a *magnetic effect* associated with it.

➤ The experiments of **Oersted** and **Ampere**, on the effects of a current-carrying wire deflecting a compass needle led to the discovery of the magnetic effects of current.

➤ Magnetic field lines due to a straight conductor carrying current:

(i) The magnetic field lines form concentric circles around the wire, with their plane perpendicular to the straight wire and with their centre lying on the wire.

(ii) If a cardboard is inserted perpendicular to the current carrying wire and some iron filings are sprinkled over the cardboard, then iron filings arrange themselves in a circular pattern.

(iii) When the direction of current in the wire is reversed, the pattern of iron filings does not change, but the direction of deflection of the compass needle gets reversed.

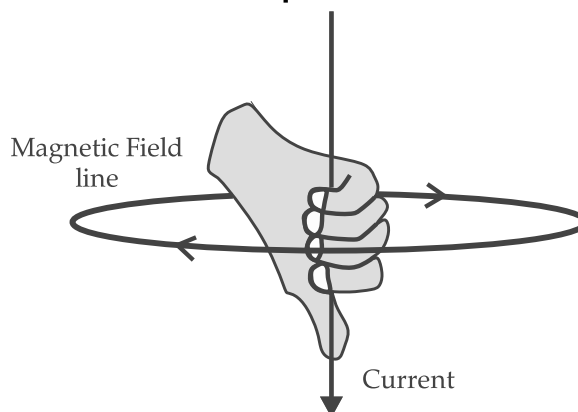
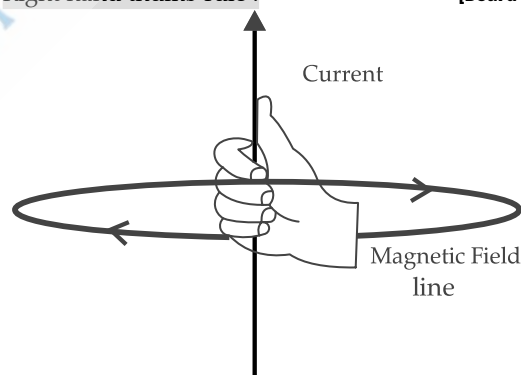
(iv) On increasing the current in the wire, the magnetic field lines become denser and the iron filings get arranged in circles up to a larger distance from the wire, showing that the magnetic field strength has increased and it is effective up to a larger distance.

[Board 2023]



➤ Right hand thumb rule :

[Board 2024]



If we hold the current carrying conductor in our right hand such that the thumb points in the direction of current, then encircling of fingers gives the direction of the magnetic field.

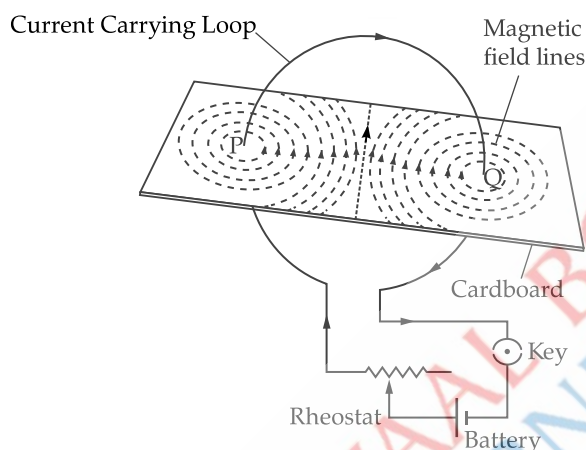
➤ It is now believed that currents (i.e., moving charges) – and not magnetic poles – are the basic cause of all magnetism.

➤ From the pattern of magnetic field lines due to a circular current carrying wire, it is noted that

(i) In the vicinity of wire at P and Q , the magnetic field lines are nearly circular.

(ii) Within the space enclosed by the wire (i.e., between P and Q), the magnetic field lines are in the same direction.

(iii) Near the centre of the loop, the magnetic field lines are nearly parallel to each other, so the magnetic field may be assumed to be nearly uniform in a small space near the centre.



Magnetic field due to current in a loop

[Board 2023,22]

(iv) At the centre, the magnetic field lines are along the axis of the loop and normal to the plane of the loop.

➤ **Comparison of Electromagnet and Permanent Magnet:**

S. No.	Electromagnet	Permanent magnet
1.	It is made of soft iron.	It is made of steel.
2.	It produces the magnetic field so long as current flows in its coils i.e., it produces temporary magnetic field.	It produces a permanent magnetic field.
3.	The magnetic field strength can be changed.	The magnetic field strength cannot be changed.
4.	The electromagnets of a very strong field can be made.	The permanent magnets are not so strong.
5.	The polarity of an electromagnet can be reversed.	The polarity of a permanent magnet cannot be reversed.
6.	It can easily be demagnetised by switching off the current.	It cannot be demagnetised easily.

➤ **Uses of Electromagnet:**

(i) It is used for lifting and transporting large masses of iron scrap, girder, plates, etc.

(ii) It is used for loading the furnace with iron.

(iii) It is used for separating the magnetic substances such as iron from debris and raw materials.

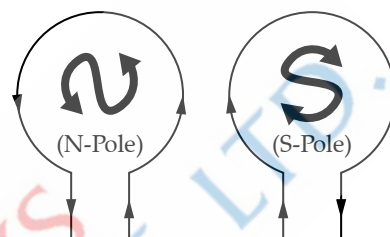
(iv) It is used for removing pieces of iron from wounds.

(v) The magnetic field lines become denser (i.e., the magnetic field strength is increased) if:

(a) the strength of the current in the loop is increased, and

(b) the number of turns in the loop is increased.

➤ For finding the polarity of the two faces of a current-carrying circular coil, the following rule applies: When an observer, looking at the circular coil, finds the current flowing in the anti-clockwise sense, that face of the coil behaves like the north-pole (N-pole) of the equivalent magnet. On the other hand, if the current is seen to flow in the clockwise sense, that face of the coil behaves like the south-pole (S-pole) of the equivalent magnet.



➤ A solenoid is made by closely winding a large number of turns of an insulated wire on a long hollow non-conducting cylinder.

➤ A current-carrying solenoid produces a **uniform field** close to its centre and its field is very similar to that of a bar magnet.

➤ When a ferromagnetic-material is placed inside a coil carrying current, it gets magnetised. Such magnets are called **electromagnets**.

➤ The electromagnet consists essentially of a soft iron core around which a large number of turns of insulated wire are wound. It is a temporary magnet which can be magnetised or demagnetised, in a very short time, simply by switching 'ON' or 'OFF' the current through it.

➤ **An electromagnet has the following advantages over a permanent magnet:**

(i) An electromagnet can produce a strong magnetic field.

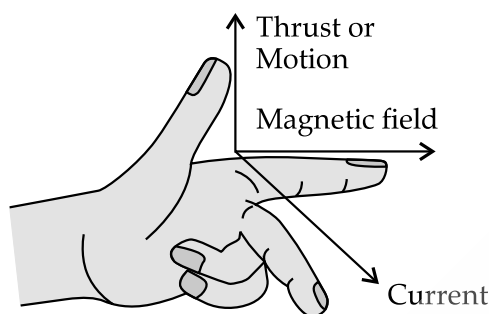
(ii) The strength of the magnetic field of an electromagnet can easily be changed by changing the current (or the number of turns) in its solenoid.

Magnetism

(iii) The polarity of the electromagnet can be reversed by reversing the direction of current in its solenoid.

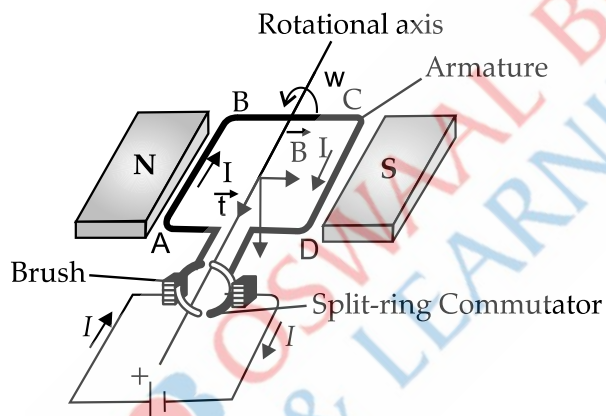
➤ When a current I passes through a conductor of length l placed in magnetic field B , then the force experienced is given by, $F = IB\sin\theta$ where θ is the angle between the length of the conductor and the magnetic field. [Board 2023]

➤ **Fleming's left hand rule:** Stretch the forefinger, central finger and the thumb of your left hand mutually perpendicular to each other. If the forefinger indicates the direction of magnetic field, central finger indicates the direction of current, then the thumb will indicate the direction of motion of conductor (i.e., force on conductor).



➤ DC motor

[Board 2024]



- Direction of rotation of motor is according to Fleming's left-hand rule.
- In motor, electrical energy is converted into mechanical energy.
- It has (a) armature coil, (b) strong electromagnet, (c) split rings and (d) carbon brush.
- When current flows through the armature coil, placed in between the poles of the magnet, it experiences a force couple, hence it rotates about the rotational axis.
- Its speed of rotation can be increased by (a) increasing number of turns, (b) increasing area of the armature, (c) increasing current and (d) increasing the magnetic field strength.



Key Words

➤ **Magnetic field lines due to a straight current carrying conductor:**

- Concentric circles around the wire.
- Lie on the plane perpendicular to the straight wire.
- Centres lying on the wire.

➤ **Right Hand Thumb Rule:**

Holding the current carrying conductor in right hand, if the thumb points in the direction of current then encircling of fingers gives the direction of the magnetic field.

➤ **Electromagnet:** The electromagnets are temporary magnets that are created when electricity passes through a metal wire coil.

➤ **Properties of electromagnet:**

- Temporary magnet.
- Generally made of soft iron.
- Magnetism disappears as soon as the flow of current through surrounding coil ceases.
- Polarity may be reversed.
- Strength may be altered.
- Field strength is greater than permanent magnet.

➤ **Force experienced by a current carrying conductor placed in magnetic field:** $F = IB\sin\theta$

where θ = angle between the length of the conductor and magnetic field.

I = current passing through a conductor

B = Magnetic field strength

L = Length of the conductor

➤ **Fleming's left hand rule:**

➤ If the forefinger, second finger and the thumb of left hand becomes mutually perpendicular to each other, then

- Forefinger indicates the direction of magnetic field
- Second finger indicates the direction of current
- Thumb indicates the direction of force on the conductor

➤ **Parts of a motor:**

- Armature
- Electromagnet
- Split rings
- Carbon brush

MNEMONICS

Concept: Fleming's left hand rule.

Mnemonics: Move Fast from Chembur to Salem For Training.

Interpretations:

M: magnetic Field

F: Fore finger

C: Current

S: Second finger

F: Force

T: Thumb

Fore finger indicates the direction of magnetic field, second finger indicates the direction of current, thumb will indicate the direction of motion of force on conductor.

TOPIC-2: ELECTROMAGNETIC INDUCTION

Concepts Covered: Electromagnetic induction, Fleming's right hand rule, AC generator, Transformer



Revision Notes

➤ Faraday's experiments were concerned with the production of an electric current from a magnetic field.

➤ From his experiments, Faraday observed that whenever there is a change in the magnetic flux (number of magnetic lines of force passing normally) linked with a circuit, an induced emf is produced. Thus, Electromagnetic Induction is the phenomenon in which an e.m.f. is induced in the coil if there is a change in the magnetic flux linked with the coil.

➤ Faraday's and Lenz's Laws on the phenomenon of electromagnetic induction are:

(i) Faraday's Law:

Whenever there is a change in magnetic flux linked with a circuit, an emf is induced. The induced emf lasts as long as the change in magnetic flux is taking place.

The magnitude of the induced emf is proportional to the time rate of change of magnetic flux.

(ii) Lenz's Law: The direction of the induced emf is such that it always opposes the change responsible for its production.

➤ From Faraday's laws the following conclusions can be drawn.

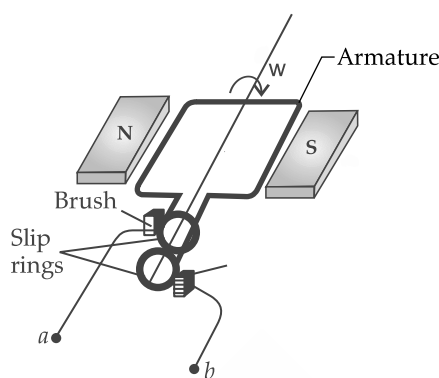
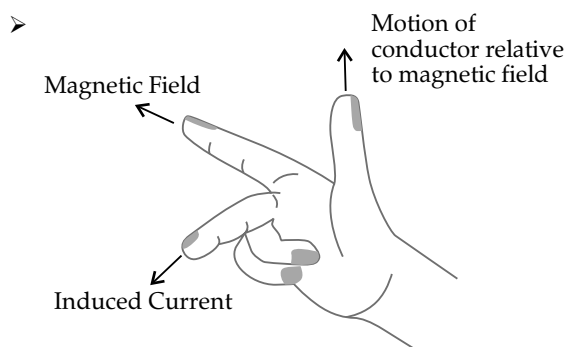
(i) A current flows in the coil when there is a relative motion between the coil and the magnet.

(ii) The direction of current is reversed if the direction of motion or polarity of the magnet is reversed.

(iii) The current in the coil may be increased (i) by the rapid motion of the magnet or coil, (ii) by the use of a strong magnet, (iii) by increasing the area of the coil and (iv) by increasing the number of turns in the coil.

➤ The direction of induced emf can be determined by Fleming's right hand rule.

➤ **Fleming's right hand rule:** It states that if the thumb, forefinger and the middle finger of right hand are spread at right angle to one another in such a way that the forefinger points in the direction of the magnetic field, thumb gives the direction of motion of conductor relative to magnetic field, then the direction in which the middle finger points, gives the direction of induced current.



➤ Advantages of AC over DC:

- AC voltage can be easily stepped up or down
- Low loss during long distance transmission
- AC generators are generally less expensive
- AC can be easily converted to DC

Example 1

Which of the following statements is incorrect?

- Whenever there is a change in magnetic flux linked with a circuit, an emf is induced.
- The induced emf lasts as long as the magnetic flux is present.
- The magnitude of the induced emf is proportional to the time rate of change of the magnetic flux.
- All of the above

Ans. Option (b) is correct.

Exp Induced emf lasts as long as the change in magnetic flux is taking place.

Example 2

The direction of induced current is obtained by:

- Fleming's left hand rule
- Fleming's right hand rule
- Right hand thumb rule
- Clock rule

Ans. Option (b) is correct.

Exp If the right hand is held with the thumb, index finger and middle finger mutually perpendicular to each other then, the thumb indicates the direction of the motion of the conductor relative to the magnetic field, the first finger indicates the direction of the magnetic field and the second finger indicates the direction of the induced current within the conductor.

AC Generator

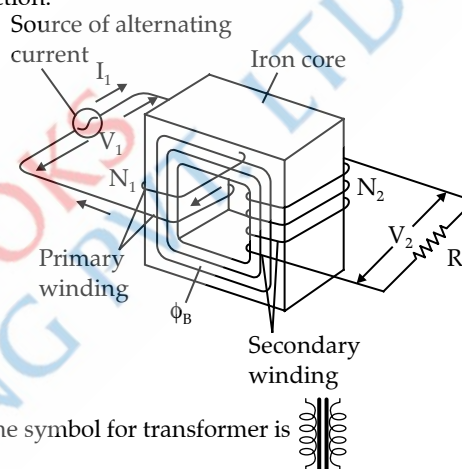
[Board 2024]

- Generator is a device which converts mechanical energy into electrical energy.
- It has (a) an armature coil, (b) slip rings, (c) carbon brushes and (d) strong magnets.
- According to Faraday's law when the magnetic flux is changed, emf is induced. According to this principle when the coil is rotated mechanically in between the poles of a magnet, emf is induced due to the change in flux and electricity is generated through the circuit.

Transformer

[Board 2024]

- It is a device which can increase or decrease the alternating voltage.
- It is based on the principle of electromagnetic induction.



- The symbol for transformer is

- It consists of two coils wound on the same core. The alternating current passing through the primary creates a continuously changing flux through the core. This changing flux induces an alternating emf in the secondary.

- (1) Transformer works on AC only and never on DC.
- (2) It can increase or decrease amplitude of AC voltage.
- (3) Transformer does not change the frequency of input AC.
- (4) There is no electrical connection between the winding but they are linked magnetically.
- (5) Let N_p = number of turns in primary, N_s = number of turns in secondary, V_p = applied (input) voltage to primary, V_s = voltage across secondary (load voltage or output), I_p = current in primary; I_s = current in secondary (or load current)

As in an ideal transformer, there is no loss of power, i.e.,

$$P_{\text{out}} = P_{\text{in}}$$

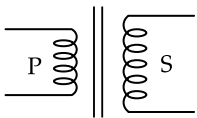
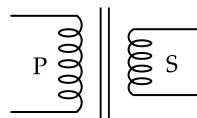
Hence,

$$V_s I_s = V_p I_p \quad \frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s} = k$$



Magnetism

k = Transformation ratio (or turns ratio)

Types of transformers	
Step up transformer	Step down transformer
It increases voltage and decreases current	It decreases voltage and increases current
	
$V_S > V_P$ $N_S > N_P$ $I_S < I_P$	$V_S < V_P$ $N_S < N_P$ $I_S > I_P$



Key Words

➤ Faraday's Law:

- Whenever there is a change in magnetic flux linked with a circuit, an emf is induced.
- The induced emf lasts as long as the change in magnetic flux is taking place.
- The magnitude of the induced emf is proportional to the time rate of change of the magnetic flux.

➤ Lenz's Law:

The direction of the induced emf is such that it always opposes the change responsible for its production.

➤ Fleming's right hand rule:

If the thumb, forefinger and the middle finger of right hand are spread at right then

- the forefinger points the direction of the magnetic field,
- thumb points the direction of motion of conductor relative to magnetic field
- middle finger points the direction of induced current.

➤ A.C. Generator:

- Converts mechanical energy into electrical energy
- It has (a) an armature coil, (b) slip rings, (c) carbon brushes and (d) strong magnets.
- It is based on the principle of electromagnetic induction.

➤ Transformer:

- Can increase or decrease the amplitude of alternating voltage.
- It is based on the principle of electromagnetic induction.
- Transformation ratio = Secondary turns/Primary turns = Secondary voltage / primary voltage

MNEMONICS

Concept: Fleming's right hand rule.

Mnemonics: Teacher listed Most Common questions of Food Management in Second Introductory Class.

Interpretation:

T: Thumb

M: Motion of

C: Conductor

F: Fore finger

M: Magnetic field

S: Second finger

I: induced

C: Current

If thumb, forefinger and the middle finger of right hand are spread at right angles, then forefinger points the direction of the magnetic field, thumb points the direction of motion of conductor relative to magnetic field and middle finger points the direction of induced current.



Heat

TOPIC-1: CALORIMETRY

Concepts Covered: Heat and temperature, Specific heat capacity, Heat capacity, Method of mixtures (calorimetric principle), Heat loss and heat gain



Revision Notes

➤ **Heat:** It is the form of energy that flows from one part to the other parts of the system due to a temperature difference. It measures the total **internal energy** possessed by the particles. It is represented by Q . The SI unit of heat is joule (J). Another unit of heat is a calorie (cal).

➤ **Temperature:** It is the measure of the **degree of hotness or coldness** of any body. Heat flows from a body at a higher temperature to a body at a lower temperature until they reach to an equal temperature. The SI unit of temperature is **Kelvin (K)**. Other units of temperature are Celsius ($^{\circ}\text{C}$) and Fahrenheit ($^{\circ}\text{F}$).

Temperature (in K) = 273 + temperature in $^{\circ}\text{C}$

Temperature in (in $^{\circ}\text{F}$) = (Temperature in $^{\circ}\text{C} \times 9/5) + 32$

$$\left[\frac{C - 0}{100 - 0} = \frac{F - 32}{212 - 32} = \frac{K - 273}{373 - 273} \right]$$

Here, C, F and K represent Celsius, Fahrenheit, and Kelvin scale.

➤ **Different units of heat energy:**

(i) **One calorie:** Calorie is the quantity of the heat needed to raise the temperature of **1 g of water from 14.5°C to 15.5°C** , i.e., through 1°C , around the mean value of 15°C or the amount of heat required to raise the temperature of 1 g of pure water by 1°C .

$$1 \text{ cal} = 4.186 \text{ J} \approx 4.2 \text{ J}$$

(ii) **One kilocalorie:** It is the amount of the heat required to raise the temperature of 1 kg of pure water by 1°C .

$$1 \text{ kcal} = 10^3 \text{ cal} = 4.2 \times 10^3 \text{ J}$$

➤ **Difference between heat and temperature:**

S. No.	Heat	Temperature
1.	The kinetic energy due to the random motion of molecules of a substance is known as its heat energy.	The quantity which determines the direction of flow of heat between the two bodies kept in contact is called temperature.
2.	The SI unit of heat is joule (J).	The SI unit of temperature is kelvin (K).
3.	It is measured by the principle of calorimetry.	It is measured by a thermometer.

➤ **Thermal capacity or heat capacity:** Thermal capacity is defined as the total amount of heat required to raise the temperature of a body by 1°C (or 1K). It is represented by c' .

[Board 2022]

$$\text{Heat capacity, } c' = \frac{\text{Heat supplied to the body (Q)}}{\text{Change in temperature } (\Delta T)}$$

Its SI unit is J°C or J/K .

Heat

➤ **Specific Heat Capacity:** It is the amount of heat energy required to raise the temperature of unit mass of the body by 1°C . Represented by c .
SI unit: $\text{J kg}^{-1}\text{K}^{-1}$

➤ **Difference between Heat capacity and Specific heat capacity:**

S. No.	Heat capacity	Specific heat capacity
1.	It is the amount of heat energy required to raise the temperature of the entire body by 1°C .	It is the amount of heat energy required to raise the temperature of unit mass of the body by 1°C .
2.	It depends both on the material and mass of the body. More the mass of the body, more is its heat capacity.	It does not depend on the mass of the body but is the characteristic property of the material of the body.
3.	Heat capacity, $c' = \frac{Q}{\Delta t}$ $= \text{mass } (m) \times \text{specific heat capacity } (c)$	Specific heat capacity, $c = \frac{Q}{m \Delta t} = \frac{\text{Heat capacity } (c')}{\text{mass } (m)}$
4.	Its unit is J K^{-1} .	Its unit is $\text{J kg}^{-1} \text{K}^{-1}$.

➤ The amount of heat gained or lost by a substance depends on the following three factors :

- (i) The rise or fall of temperature of the substance.
- (ii) The quantity or the mass of the substance.
- (iii) The nature of substance.

➤ The supply of heat to a body generally increases the kinetic energy of its molecules, and this shows itself as a rise in the body's temperature.

➤ Relation between thermal capacity and specific heat capacity:

Thermal capacity = mass $\times$ specific heat capacity.

➤ We may view the specific heat capacity as one of the factors on which the thermal inertia of the body depends. More the specific heat capacity, the more it is difficult to change the thermal state, i.e., the temperature of the body.

➤ Specific heat capacity of water = $1 \text{ cal}/(\text{g}^{\circ}\text{C}) = 4.2 \times 10^3 \text{ J}/(\text{kg}^{\circ}\text{C})$.

➤ The specific heat capacity of water is much higher than that of most substances. Thus, water has a high thermal inertia.

➤ According to the principle of heat transfer or principle of calorimetry :

➤ **SI units:**

S. No	Physical quantity	SI unit	
		Name	Symbol
1.	Specific heat capacity	Joule per kilogram kelvin	$\text{J}/(\text{kg K})$
2.	Thermal capacity	Joule per kelvin	J/K
3.	Heat	Joule	J

➤ **Relevant physical constants :**

(1) Specific heat capacity of water = $1 \text{ cal}/(\text{g}^{\circ}\text{C}) = 4.2 \times 10^3 \text{ J}/(\text{kg}^{\circ}\text{C})$

S. No	Substance	Sp. heat capacity	
		in $\text{J/kg}^{\circ}\text{C}$	in $\text{cal}/(\text{g}^{\circ}\text{C})$
1.	Aluminium	882	0.21
2.	Copper	399	0.095
3.	Gold	134	0.032
4.	Iron	483	0.115
5.	Lead	130	0.031
6.	Brass	386	0.092
7.	Silver	235	0.056

The heat gained by the cold body is equal to the heat lost by the hot body, provided no transfer of heat to the surroundings.

➤ When m kg of a substance of specific heat capacity c is heated (or cooled) through $\theta^{\circ}\text{C}$, the quantity of heat Q gained (or lost) by the substance is given by

$$Q = mc\theta$$

➤ The equilibrium temperature θ , during heat transfer, is given by

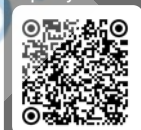
$$m_1c_1(\theta_1 - \theta) = m_2c_2(\theta - \theta_2)$$

$$\text{Or } \theta = \frac{[m_1c_1\theta_1 + m_2c_2\theta_2]}{[m_1c_1 + m_2c_2]}$$

➤ **The method of mixtures** is based on the principle of heat transfer is often used for the measurement of the specific heat capacity of a given body.

➤ The high value of the specific heat capacity of water finds many applications in nature, and in practice. For example, due to the high specific heat of water, it is used as a coolant.

Applications of High Specific Heat Capacity of Water



Scan Me!

(2) Specific heat capacity of ice = $2100 \text{ J kg}^{-1} \text{K}^{-1}$

(3) The following table gives the values of Specific heat capacities of some common substances :

S. No	Substance	Sp. heat capacity	
		in $\text{J/kg}^{\circ}\text{C}$	in $\text{cal}/(\text{g}^{\circ}\text{C})$
8.	Zinc	291	0.0693
9.	Tin	231	0.055
10.	Alcohol	218	0.52
11.	Kerosene	210	0.50
12.	Turpentine	176	0.42
13.	Mercury	139	0.033
14.	Water	4200	1.00

Example 1

The specific heat capacity of substance A is $3,800 \text{ J kg}^{-1} \text{ K}^{-1}$, and that of substance B is $400 \text{ J kg}^{-1} \text{ K}^{-1}$. Which of the two substances is a good conductor of heat? Give a reason for your answer.

Ans. Substance B is a good conductor of heat because the specific heat capacity of B is less than that of A. Specific heat capacity is the amount of heat energy required to raise the temperature of 1 kg of a substance by 1K. So, the substance B gets heated faster. [ICSE Marking Scheme, 2019]

Example 2

(i) State whether the specific heat capacity of a substance remains the same when its state changes from solid to liquid.

(ii) Give one example to support your answer.

Ans. (i) No, the specific heat capacity of a substance does not remain same when its state changes.

(ii) Specific heat capacity of ice = $2.1 \text{ J/g } ^\circ\text{C}$ and that of water is $4.2 \text{ J/g } ^\circ\text{C}$. [ICSE Marking Scheme, 2019]

MNEMONICS

Concept: Quantity of the heat required to raise the temperature of a body (Q) = $mc\Delta t$.

Mnemonics: Quiz Master Contributed our Team.

Interpretation:

Q: Quantity of heat

M: Mass of the body

C: Specific heat capacity

T: Temperature difference

Quantity of heat required to raise the temperature of a body (Q) = mass $\times$ specific heat capacity $\times$ temperature difference

$$Q = mc\Delta T$$

**Key Words**

- **Heat:** It is the form of energy.
- **Temperature:** Parameter that tells the thermal state of a body. Determines the direction of heat flow.
- **Thermal capacity or heat capacity:** The total amount of heat required to raise the temperature of a body by 1°C (or 1K).
- **Specific Heat Capacity:** Amount of heat energy required to raise the temperature of unit mass of the body by 1°C .
- **Principle of calorimetry (Principle of mixtures):** The heat gained by the cold body is equal to the heat lost by the hot body.
- **The equilibrium temperature θ ,** during heat transfer, is given by

$$\theta = \frac{m_1 c_1 \theta_1 + m_2 c_2 \theta_2}{m_1 c_1 + m_2 c_2}$$

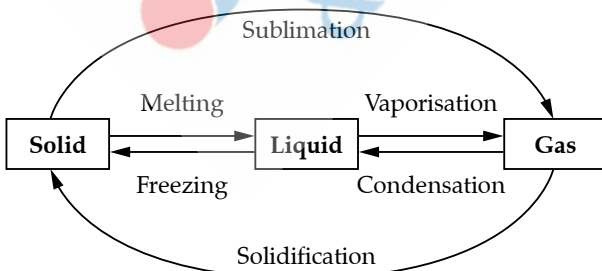
TOPIC-2: LATENT HEAT

Concepts Covered: Change of phase (state), Latent heat, Heating curve for water



Revision Notes

➤ Phase change diagram:



➤ The phenomenon of melting, freezing, boiling (evaporation) and condensation make up the subject matter of change of state.

➤ **During the change of state, average kinetic energy does not change but average potential energy changes (increases on melting).**

➤ The melting or fusion of crystalline solid into its corresponding liquid takes place (under given external

condition) at a particular temperature known as its melting point.

➤ The **specific latent heat of fusion** of a given solid is the heat required to melt 1 kilogram of it, at its melting point, without change in temperature. It is measured in J/kg.

[Board 2024,22]

➤ **Specific latent heat of fusion of ice** = 336×10^3 J/kg.

➤ The heat Q needed to melt m kg of a substance of specific latent heat of fusion L , is given by $Q = mL$.

➤ The boiling point of a liquid, under the given external conditions, is that constant temperature at which the liquid starts boiling, i.e., gets converted into the vapour state (at a very fast rate).

➤ We can explain a number of natural phenomena using the idea of specific latent heat of fusion.



Heat

- In cold countries, water in lakes does not freeze all at once.
- Drinks get more cool by adding ice than by being ice-cold water at 0°C .
- It is colder after a hail storm than before or during it.

➤ Change of state curve

[Board 2023,22]

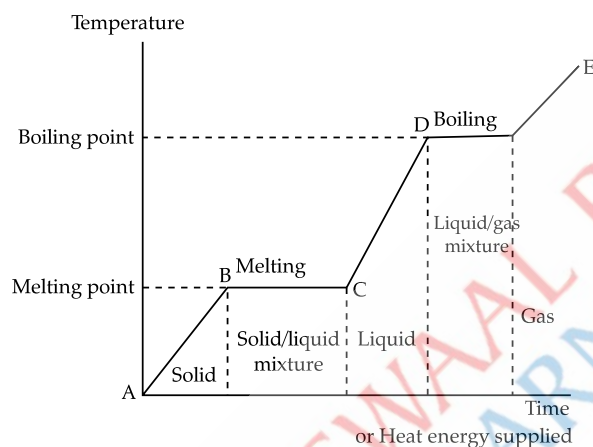
When a substance is heated, its temperature rises.

When the temperature reaches its melting point, the temperature becomes constant till the total solid mass is converted into liquid. During this stage, the solid substance absorbs latent heat and increases the potential energy of the molecules so that a change of state takes place.

After full conversion into liquid, the temperature again starts increasing till it reaches its boiling point where again the temperature becomes constant till the whole liquid mass is converted into gaseous state.

After full conversion into a gaseous state, the temperature again starts rising.

Graphically, this can be represented by plotting temperature along Y axis and time along X axis.



Since, specific heat capacity of liquid is more than the specific heat capacity of solid, therefore the slope of CD is less compared to the slope of AB portion.

MNEMONICS

Concept: Liquid state to gaseous state conversion is vaporisation. Liquid state to solid state conversion is freezing.

Mnemonics: Little Guy Vanished in the Large Space of the Fair.

Interpretations:

- L: Liquid to
- G: Gaseous state
- V: Vaporisation
- L: Liquid to
- S: Solid
- F: Freezing

MNEMONICS

Concept: Solid state to liquid state conversion is melting. Gaseous state to liquid state conversion is condensation.

Mnemonics: Mina Slept Late. She Came to the Gathering Late.

Interpretations:

- M: Melting
- S: Solid to
- L: Liquid
- C: Condensation
- G: Gaseous state
- L: Liquid



Key Words

- **Melting (fusion):** Solid state to liquid state conversion
- **Freezing:** Liquid state to solid state conversion
- **Vaporisation:** Liquid state to gaseous state conversion
- **Condensation:** Gaseous state to liquid state conversion
- **Sublimation:** Solid state to gaseous state conversion
- **Solidification:** Gaseous state to solid state conversion
- **Specific latent heat of fusion:** Heat required to melt 1 kilogram of solid, at its melting point, without change in temperature.



Modern Physics



Revision Notes

➤ **Atomic number:** The atomic number (Z) of an atom is the number of protons in its nucleus (which is the same as the number of electrons in a neutral atom).

➤ **Mass number:** The mass number (A) of an atom is the total number of nucleons (i.e., number of protons and neutrons combined) in its nucleus.

➤ **Isotopes:** The atoms belonging to the same element, having same atomic number Z , but different mass number A , are called isotopes.

Examples include: ${}^1_1\text{H}$, ${}^2_1\text{H}$ and ${}^3_1\text{H}$ which are the three isotopes of hydrogen.

➤ **Isotones:** The atoms having a different number of protons but same number of neutrons, i.e., different Z and A , but same $A - Z$, are called isotones. They have a different number of electrons.

Examples of isotones include: ${}^{23}_{11}\text{Na}$ and ${}^{24}_{12}\text{Mg}$

➤ **Isobars:** The atoms of different elements which have the same mass number A , but a different atomic number (Z), are called isobars.

Examples of isobars include: ${}^{23}_{11}\text{Na}$ and ${}^{23}_{12}\text{Mg}$

➤ The substances which disintegrate (or decay) by the spontaneous emission of radiation, are called radioactive substances, for example, uranium, radium, polonium, thorium, actinium, etc.

➤ **Radioactivity** is the process of spontaneous decay of unstable nuclei emitting α or β particles and γ rays from the nucleus of atoms.

➤ **Effect on atomic number (Z) and mass number (A) due to alpha, beta and gamma emissions:**

Quantity	α emission	β emission	γ emission
(i) Z	Decreases by 2	Increases by 1	No change
(ii) A	Decreases by 4	No change	No change

➤ **Properties of α -particles:**

[Board, 2024]

➤ α -particle consists of a helium nucleus. It is denoted by ${}^4_2\text{He}$ (atomic number 2 and mass number 4). It is a positively charged particle.

➤ It has least penetration power and highest ionisation power.

➤ They can't penetrate the skin but is harmful to the human body.

➤ α -particles effect a photographic plate.

➤ α -particles cause fluorescence on striking a fluorescent material.

➤ These are affected by electric and magnetic fields.

➤ **Properties of β -particles :**

[Board, 2024]

➤ β -particles are highly energetic electrons, which are released from inside of a nucleus. It is denoted by ${}_{-1}\beta^0$ (atomic number -1 and mass number 0) and ${}_{+1}\beta^0$ (atomic number 1 and mass number 0).

➤ β -particles may be either negatively or positively charged, and it has negligible mass.

➤ For the emission of negative β -particle, a neutron changes into a proton and an electron, inside the nucleus and for $+\beta$ -particle, a proton converts into neutron and a positron.



- β -particle has 100 times penetration power as compared with α -particle.
- It has less ionisation power but is still harmful.
- β -particles effect a photographic plate.
- β -particles cause fluorescence on striking a fluorescent material.
- These are affected by electric and magnetic fields.

[Board, 2023]

➤ **Properties of γ -rays:**

- High frequency wave which don't have mass and moves with speed of light.

➤ **Difference between α , β and γ radiations:**

Characteristic	α -particle	β -particle	γ -particle
Compositions	A helium nuclei consisting of two protons and two neutrons	High-energy electron	High-energy electromagnetic radiation
Charge	Positive and twice to that of a proton $(+2e) = +3.2 \times 10^{-19}C$	Negative, equal to that of an electron $(-e) = 1.6 \times 10^{-19}C$	Neutral, no charge
Mass	Four times the mass of a proton $= 6.68 \times 10^{-27} \text{ kg}$	Same as the mass of an electron $= 9.1 \times 10^{-31} \text{ kg}$	No mass
Speed	About 3.33% the speed of light $= 10^7 \text{ ms}^{-1}$	About 90% the speed of light $= 27 \times 10^7 \text{ ms}^{-1}$	Same as the speed of light $= 3 \times 10^8 \text{ ms}^{-1}$
Magnetic and electric field effect	Deflected towards negative plate or terminal	Deflection towards positive plate or terminal	No deflection
Relative angle of deflection	Less	More	Zero
Penetration power	Small, 2.7–8.0 cm in air	Large, 5 m in air	Very large, 500 m in air
Absorbed or stopped	Thin cardboard, human skin	5 mm thick aluminium sheet, 1 mm thick lead	30 cm thick iron sheet, 10 cm thick lead
Ionising power	Highest 10^4 times that of γ radiations and 10^2 times that of β -particles	Moderate, 100 times less than α -particles but 100 time more than γ -radiation	Low, 10^{-4} times of β -particles and 10^{-2} times that of α -particles
Biological damage	Less damage	More damage	Immense damage

- **Radioisotopes** are the unstable form of an element that emit radiation to transform into a more stable form.

➤ **Uses of Radioisotopes:**

- Medical use: In treatment of leukemia, cancer
- Scientific use: Carbon dating
- Industrial use: As fuel for atomic energy reactors
- Sources of harmful radiations are radioactive fallout from nuclear power plants, nuclear waste and other sources (cosmic radiations, X-rays).
- Harmful effects of radiation include short-term effects (diarrhoea, sore throat, loss of hair), long-term effects (leukemia, cancer) and genetic defects.
- **Safety measures to be taken while using nuclear energy:**

• **While establishing a nuclear power plant:** Nuclear plant must have a shield with lead and steel walls, housing it in an airtight building with a backup of a cooling system.

• **While handling radioisotopes:** People handling radioisotopes should put on lead-lined aprons and gloves and handle the material with long lead tongs.

- It is neutral in nature and denoted by ${}^0_0\gamma$.
- It has 1000 times and 100 times penetration power as compared with α - and β -particles, respectively. It is very harmful to the living cell.
- These radiations affect a photographic plate.
- These radiations cause fluorescence when they strike a fluorescent material.
- These rays are not affected by electric and magnetic fields.

• **In safe disposal of nuclear waste:** Nuclear waste must be kept in thick casks and must be buried in specially constructed deep underground stores away from populated areas.

➤ Background radiations are the radioactive radiations, which exist even in the absence of actual radioactive source. These are emitted both by internal and external sources.

➤ **Radioactive fallout:** An air burst can produce minimal fallout if the fireball does not touch the ground. On the other hand, a nuclear explosion occurring at or near the earth's surface can result in severe contamination by the radioactive fallout. Radioactive fall outs are the airborne radioactive particles that fall to the ground during and after an atomic bombing, nuclear weapons test or nuclear plant accident. It has been studied that radioactive fallout is not only harmful to animal world directly, indirectly it also contaminates agricultural land, surface-soil layer and surface-water.

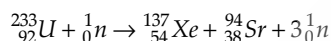
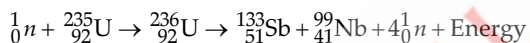
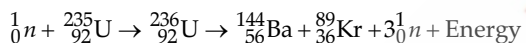
➤ **X-rays and background radiation:** Some imaging procedures use X-rays, which are a special type of

radiation called 'ionising radiation.' Ionising radiation is also found all around us in the soil, sun, air, plants, food, drink and buildings. This is called background radiation. Ionising radiation may cause damage to the cells in our body. This is usually very minor and does not cause any serious damage, however, large doses may cause the cells to become cancerous. A very low dose X-ray, such as a chest X-ray, has a tiny risk. CT scans, which use higher doses of X-rays, have a higher risk.

➤ **Chemical reaction and nuclear reaction:** Chemical reaction is the formation of a new compound molecule due to rearrangement of valence electrons in interacting atoms and molecules. In this process, the nucleus and the electrons in the inner orbits remain unaffected. Energy involvement is in eV order. In nuclear reactions, the nuclei, not electrons, of the reactants interact with each other. They result in the formation of new elements. In nuclear reactions energy involvement is in MeV order.

➤ **Nuclear fission:** Beyond natural radioactive decays when a nucleus is bombarded with other nuclear particles such as proton, neutron, α -particle, etc., radioactive daughter nuclei are produced. The most important neutron-induced nuclear reaction is called fission.

When a uranium isotope ${}^{235}_{92}\text{U}$ is bombarded with a neutron it splits into two intermediate mass nuclear fragments. Three different sets of fragments may be produced:



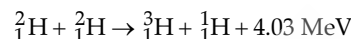
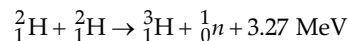
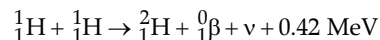
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About 200 MeV energy is produced per fission. This enormous amount of energy generated in nuclear reactors is utilised to produce electricity and the same

amount of energy is utilised in atom bomb as uncontrolled nuclear fission for destructive purpose.

➤ **Nuclear fusion:** Nuclear fusion is a nuclear process in which two light nuclei fuse to form a larger nucleus and energy is released.

Some examples of such energy liberating nuclear fusion reactions are:



MNEMONICS

Concept: Isotope, Isotone and Isobar

Mnemonics: Topewala Sales Pista, Tonebabu

Sales Nasta, Barbeque Nothing Sasta.

Interpretation:

Tope: Isotope

S: Same

P: Proton

Tone: Isotone

S: Same

N: Neutron

Bar: Isobar

N: Nucleons

S: Same

Isotope has same number of protons and different number of neutrons.

Isotone has same number of neutrons and different number of protons.

Isobar has neither same number of protons nor same number of neutrons. But total number of nucleons are same.